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Applications of electrochemical sensors for determining acetaminophen and ibuprofen in water samples: A review

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ABSTRACT

Acetaminophen (APAP) and ibuprofen (IBU) are among the most widely used analgesics for the treatment of fever, pain, and inflammation. While they play a crucial role in improving health conditions, their extensive use, often without the need for a medical prescription, has led to growing environmental concerns. The review highlights recent advances from 2019 to 2024 in the development of electrochemical sensors for the detection of APAP and IBU in various water matrices, including surface water, groundwater, wastewater, tap water and bottled water. In addition, it provides an overview of their metabolic pathways and electrochemical oxidation mechanisms. Materials used for electrode modification, such as carbon-based nanomaterials, metallic nanoparticles, metal-organic frameworks (MOFs), and synthetic polymers, have led to substantial improvements in sensor sensitivity and selectivity. The electrochemical detection of APAP and IBU offers a promising strategy for accurate and reliable analysis, with significant implications for environmental monitoring and public health. This review presents a comprehensive evaluation of the analytical performance of various sensors, comparing key analytical parameters such as limit of detection (LOD), linearity range, and relative standard deviations (RSD). The review also outlines key future research directions, including the use of biosensors and automation of detection systems to increase to increase their applicability directly in water environments.

1. Introduction

In the period from 2019 to 2024 has been increasing attention from researchers on the presence of pharmaceutical compounds in the environment, particularly in aquatic systems [1]. This is primarily due to the significant environmental risks these substances may pose to aquatic organisms [2–4]. Among these pharmaceuticals, analgesics, alongside antibiotics, are of particular concern as they are widely consumed daily, often without restrictions, by the general population. In 2023, the global market volume of acetaminophen was approximately 145,000 tons [5], and for ibuprofen was 46,000 tons [6]. These substances are also considered pseudo persistent contaminants. Fig 1 illustrates the distribution pathway of these compounds in the environment.

The process begins with pharmaceutical companies and continues with the distribution of these compounds to the population through pharmacies and hospitals. From there, they enter wastewater systems [7], where conventional treatment methods may not completely remove them, as will be discussed later. After treatment, residual compounds are discharged into surface water bodies [8] such as lakes, rivers, and seas, with some infiltrating into groundwater sources [9]. Water from these sources is subsequently treated for drinking purposes, and studies have reported the presence of pharmaceuticals residues in drinking water [10]. The persistence of these compounds in natural waters can also have adverse effects on aquatic organisms [11,12].

Acetaminophen, chemically known as N-acetyl-aminophenol (APAP) is one of the principal representatives of the analgesic class. It is also classified as an antipyretic drug and is commonly referred to as paracetamol. The World Health Organization (WHO) has included APAP in its list of essential medicines, recognizing it as one of the most effective, safe, and affordable drugs for treating priority health conditions [13]. It is a non-opioid medication used to relieve pain and reduce fever [14]. Another equally important representative of this class is (RS)-2-(4-(2-methylpropyl) phenyl) propanoic acid, commonly known as Ibuprofen (IBU). It is the third most widely used non-prescription, non-steroidal drug globally, recognized for its anti-inflammatory, analgesic, and antipyretic properties. IBU is commonly used to treat

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conditions such as fever, migraine, muscle pain, rheumatoid arthritis, toothaches, and osteoarthritis [15]. It has been shown to induce oxidative stress, cellular toxicity, genetic damage, and neurotoxic effects, including behavioral changes in aquatic organisms [16]. IBU is listed among the top 10 priority pharmaceuticals (Nonsteroidal Anti-Inflammatory Drugs, NSAIDs) that the European Union recommends for monitoring in aquatic ecosystems [17,18].

Analgesic compounds are harmful to water environments [19] for several reasons. Due to their complex molecular structure, they exhibit high stability and are not easily removed by conventional wastewater treatment processes. They are compounds with high solubility, with $LogK_{ow}$ values of 3.5 for IBU and 0.475 for APAP [20] (K_{ow} = Octanol-water partition coefficient). Furthermore, since they are not volatile substances, they cannot be removed through volatilization processes [21]. These compounds are usually found in low concentrations, which makes their removal even more challenging [22]. Wastewater treatment plants are primarily designed to eliminate biological pathogens and general organic matter: as such, conventional technologies are often insufficient for removing trace level pharmaceuticals. Advanced treatment strategies used to remove these contaminants include oxidation processes [23,24], ultrasonic irradiation [25], electrocoagulation [26], and biological removal methods [27]. Although these methods show high efficiency, they suffer from several disadvantages, such as poor selectivity, high consumption of hazardous reagents, slow processing times, and the formation of secondary substances, such as the chlorinated derivatives, that are even more hazardous [28,29]. Incomplete disposal of APAP and IBU can cause bioaccumulation and ecological damage [30,31]. As a result, these pharmaceutical compounds can easily migrate to other water sources, including both surface and groundwater.

Several analytical techniques are used for the determination of APAP/IBU in water samples including spectrophotometric methods [32], chromatographic methods [33–36], and mass spectrometry [37]. While conventional analytical techniques mentioned above remain effective, interest in electrochemical methods has increased due to their operational simplicity and potential for on-site monitoring [38–42]. When compared to traditional analytical methods, which are often more expensive, more difficult to use, time-consuming, and involve extensive sample preparation, electrochemical methods offer a cost-effective and more efficient alternative. Table 1 provides more information about the advantages and disadvantages of each method.

*HPLC-DAD (High-Performance Liquid Chromatography with Diode Array Detection), GC–MS (Gas Chromatography with Mass Spectrometry), and LC-MS (Liquid Chromatography with Mass Spectrometry)

This review aims to highlight the advantages of electrochemical methods for the determination of acetaminophen and ibuprofen in water samples only, using different sensors, and to provide a comparison of their effectiveness and, Fig 2 schematically presents the structures of the most used modifiers for this type of analysis, along with the setup of an electrochemical workstation. A wide variety of materials have been used to improve the performance of electrochemical sensors. Among the most widely used are carbon-based nanomaterials, such as carbon nanotubes (CNTs), including both single-walled (SWCNT) and multi-walled (MWCNT) types, and graphene oxide (GO). Metals and metal oxide nanoparticles, such as gold (Au), silver (Ag), and iron oxide (Fe_3O_4), are also frequently used due to their excellent catalytic and electrical properties. In addition, conductive polymers such as polypyrrole (PPy), Nafion, and chitosan, as well as advanced materials such as metal-organic frameworks (MOFs), have shown significant promise in improving the sensitivity, selectivity, and stability of sensors. It should be noted that technology in this field is advancing rapidly: therefore, the review focuses on sensors developed within the last five years (2019 –

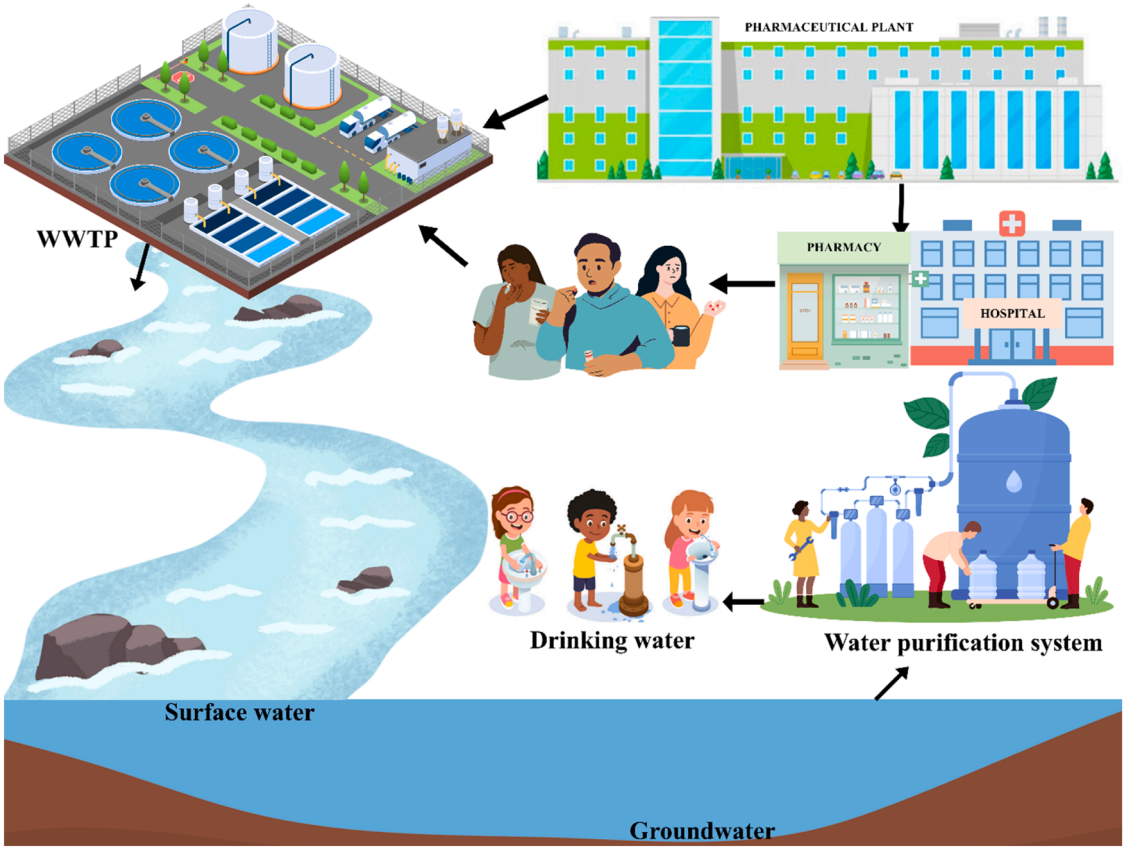


Fig. 1. Schematic representation that illustrates the pathway of acetaminophen and ibuprofen, starting from the production process to their distribution among the population through hospitals and pharmacies.

Table 1
A brief overview of the advantages and disadvantages of other analytical methods
*The references about the prices: spectrophotometric methods [43], chromatographic methods [44], mass spectrometric methods [45], and electrochemical methods [46].

Methods	Advantages	Disadvantages
Spectrophotometric	1. Rapid results 2. Wide range of applications 3. High sensitivity and accuracy 4. Non-destructive nature of measurement 5. Qualitative and quantitative analysis	1. Sample preparation 2. Sensitivity to lighting conditions 3. Relatively high LOD, which may limit their effectiveness in detecting trace concentrations in water samples 4. Expensive apparatuses, a simple spectrophotometer can cost around $2000-$8000 while advanced ones can go up to $30,000
Chromatographic	1. High sensitivity, can detect ultra-trace levels 2. Require a small amount of sample for analysis 3. Can eliminate matrix interferences 4. Have a wide application in different types of samples	1. Not simple for routine analysis 2. Sample preparation 3. It is necessary to couple it with other spectrometric apparatus, such as HPLC-DAD, GC–MS, LC-MS 4. Some methods use organic solvents, not environmentally friendly 5. Expensive instruments, they range from $30,000 to $100,000 or more
Mass spectrometry	1. It is a sensitive, quick, and accurate method 2. Requires a smaller amount of sample 3. It is a selective method	1. Mass spectrometry is usually coupled with a chromatographic device 2. Requires a qualified technician 3. Usually, the sample is destroyed during the analysis 4. Expensive equipment, typically ranges from $50,000 to $500,000 or more
Electrochemical	1. It is a fast, accurate, and easy-to-use method 2. Require minimal reagents 3. Minimal sample preparation 4. It can be portable, making them suitable for on-site and real-time analysis 5. Low cost, the price of the equipment ranges from $1000 to more advanced instruments priced at $30,000	1. Limited selectivity and sensitivity without electrode modifiers 2. Contamination of the electrode can cause surface passivation and reduced sensitivity. 3. Overlapping peaks can occur in matrices containing multiple electroactive compounds with similar structures.

2024) to ensure relevance and accuracy. This time frame may also inspire new opportunities for researchers in the field.

Many reviews have focused on the determination of acetaminophen and ibuprofen in human biological samples such as urine, serum, oral fluids, and blood [47–49], where accumulation may result from exceeding the recommended daily intake levels of these substances [50,51]. However, it should be emphasized that contamination of water sources with these compounds poses a significant risk to children, the elderly [52] and especially to individuals with chronic liver and kidney diseases [53,54]. In such patients, the adverse effects on these organs are especially pronounced. This work aims to provide an overview of the electrochemical sensors that have been used to detect APAP and IBU in water.

2. Metabolic pathways and fundamental electrochemical properties of acetaminophen and ibuprofen

APAP belongs to the class of phenolic compounds and is derived from 4-aminophenol, in which one hydrogen atom of the amino group is substituted with an acetyl group. According to the Biopharmaceutics Classification System (BCS), it is classified as Class I drug, characterized by high solubility and high permeability [55]. In living organisms, the majority of APAP is metabolized in the liver into two non-toxic compounds: the glucuronide conjugate (APAP-GLUC, 47 – 62 %) and the sulfate conjugate (APAP-SULF, 25 – 36 %) which are subsequently excreted in the urine [56]. About 5 % of APAP is excreted in its unchanged form in the urine [57], directly entering the aquatic ecosystem. Between 5 % and 10 % of APAP is oxidized by the liver cytochrome P450 enzyme system to form N-acetyl-p-benzoquinone imine (NAPQI) [58]. This compound is normally detoxified through conjugation with glutathione, a reaction catalyzed by glucuronyl S-transferase in the liver, resulting in the formation of NAPQI-GSH, a non-toxic product, that is excreted in the urine as mercapturic acid and cysteine conjugates [59], (see Fig 3A). However, this detoxification pathway can fail when glutathione stores are depleted and the amount of APAP exceeds the liver's detoxification capacity, potentially leading to hapatic injury and the accumulation of free NAPQI [60].

IBU is a non-steroidal anti-inflammatory drug (NSAID) derived from propionic acid. It is classified as a Class II drug in the BSC, exhibiting low solubility at lower pH [61]. IBU is almost entirely metabolized in the body by cytochrome P450 enzymes, primarily into non-toxic hydroxy and carboxyl metabolites (see Fig 3B), which can be excreted via urine [62]. About 1 % of IBU is excreted directly in its pure form [63]. Ibuprofen (IBU) can also be metabolized in living organisms into active metabolites, such as acyl glucuronides [64], which have been shown to be more toxic than IBU itself. Covalent modification of biomacromolecules, such as proteins or DNA, by these reactive metabolites can trigger immune responses and lead to cellular dysfunction [65]. The complete metabolic pathways of both APAP and IBU are illustrated in Fig 3.

Healy et al., [66] studied the oxidation process of APAP using cyclic voltammetry (CV) with a glassy carbon electrode (GCE) modified with β-cyclodextrin (β-CD) in phosphate buffer electrolyte. In the resulting voltammogram (see Fig 4A), a redox couple can be observed. APAP undergoes electrochemical oxidation via $2e^-/2H^+$ mechanism, leading to the formation of NAPQI as the oxidized product [67]. This oxidized species can be reduced at more negative potentials, exhibiting either reversible or quasi-reversible behavior [68]. β-CD has been utilized as an electropolymerizing agent [69] on the GCE surface, resulting in APAP oxidation. Additionally, Gharous et al., [70] investigate the oxidation of APAP, and both studies confirmed that the oxidation process is sensitive to pH changes in the solution (see Fig 4B), which influence both the peak potential and the peak current intensity.

The oxidation process of IBU is not as well-defined as that of APAP. This is largely due to its more complex molecular structure, which contains only a carboxylic group as its primary functional group, representing the highest oxidation state of carbon in organic compounds [73]. Two main aspects have been proposed by authors to explain the electrochemical oxidation of ibuprofen and why its electrochemical detection is possible, despite earlier studies that considered ibuprofen to be electrochemically inactive [74].

Prete et al., [71] investigated the oxidation mechanism of IBU using boron-doped ultrananocrystalline diamond (BD-UNCD) electrode and analyzed it via cyclic voltametric. The results were consistent with previous findings [75,76], showing that IBU exhibits a single oxidation peak (see Fig 4C). The process involves an initial electron transfer, leading to the formation of a radical cation, followed by a decarboxylation step. Notably, the peal position is not influenced by pH [77] (see Fig 4D). A similar electrochemical mechanism has been proposed for naproxen [78], due to the structural similarities between the two

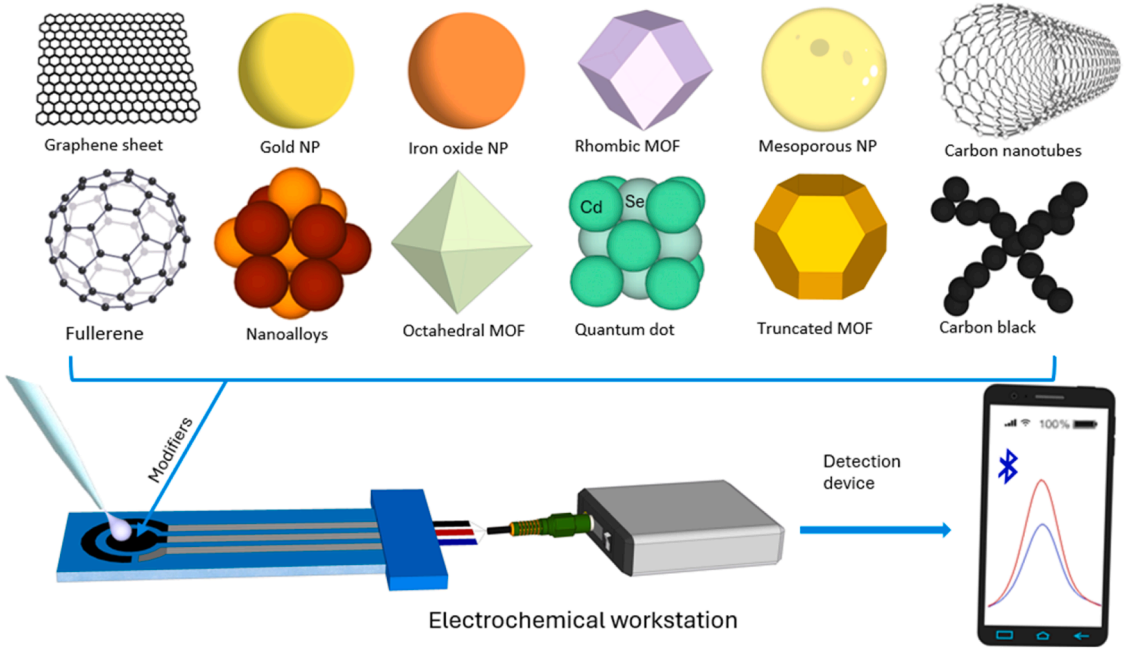


Fig. 2. Schematic representation of an electrochemical sensor along with the structures of the most used compounds as surface modifiers.

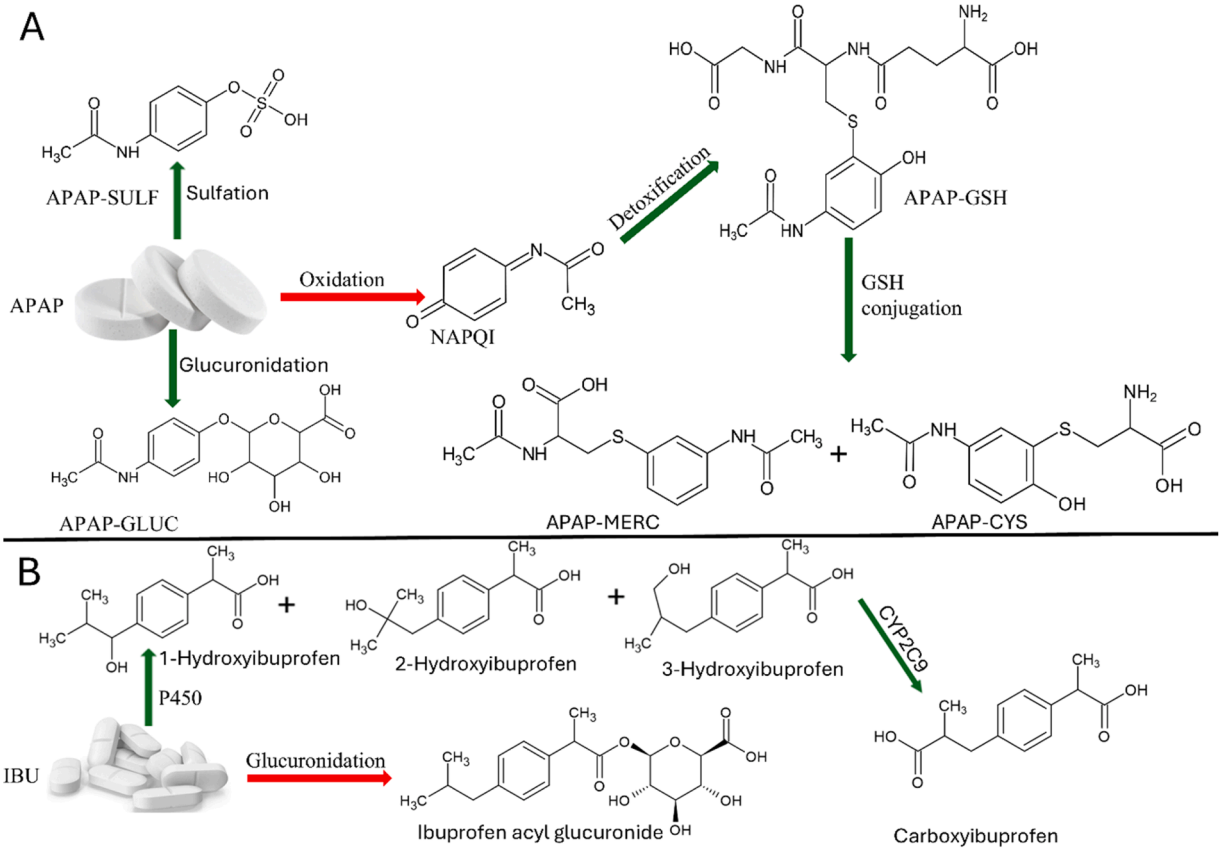


Fig. 3. The cycle of metabolism in a living organism. A) APAP, B) IBU. The red arrows indicate toxic compounds, and the green arrows indicate non-toxic compounds.

compounds. In another paper [79], it is suggested that the oxidation process of IBU is an adsorption-controlled process.

3. Electrochemical sensors fabricated by carbon compounds

Carbon-based nanomaterials have become key components in the design of high-performance electrochemical sensors due to their

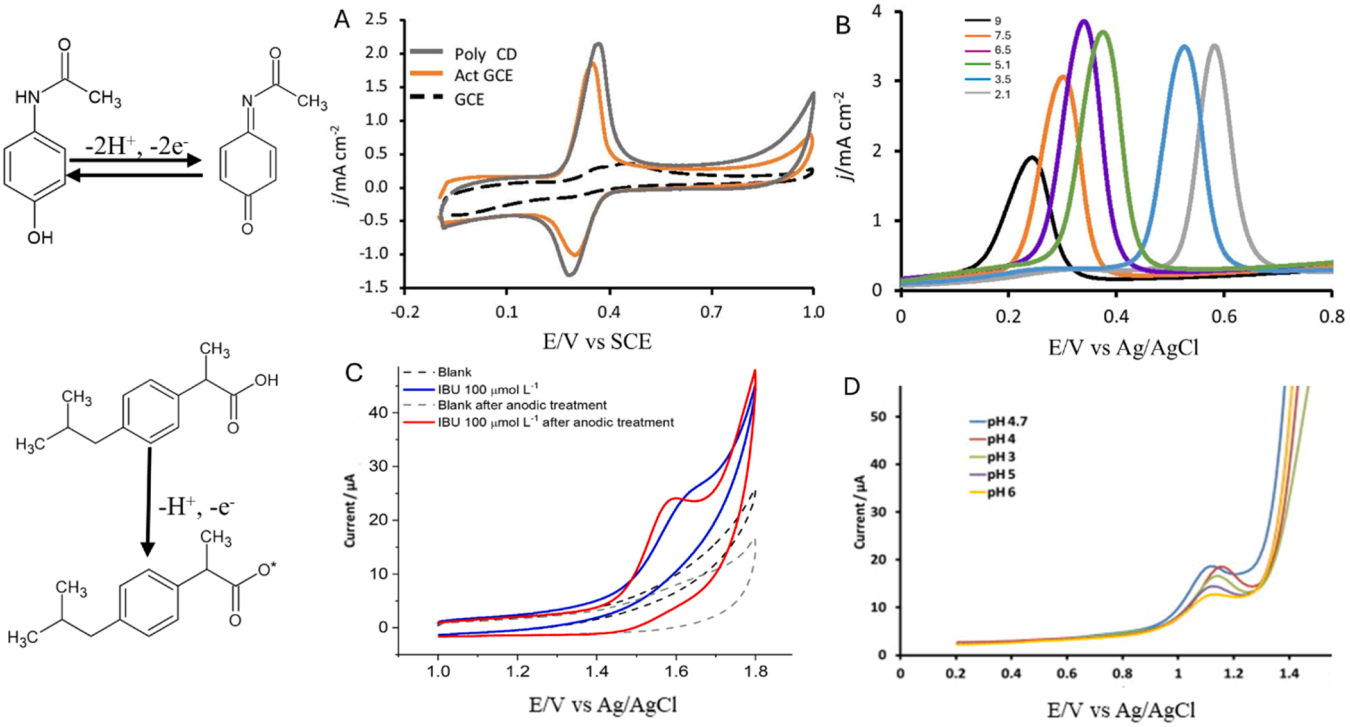


Fig. 4. Schematic representation of voltammetric responses based on data from the two previously mentioned articles. (4A) Cyclic voltammogram of APAP; (4B) Differential pulse voltammogram of APAP recorded at different pH values; (4C) Cyclic voltammogram of IBU; (4D) Square wave voltammograms of IBU in different pH. Adapted with permission from references [66,71,72].

exceptional physicochemical properties, including high electrical conductivity, large specific surface area, chemical stability, and mechanical strength [80]. These materials, which exist in various allotropic forms such as nanotubes, nanofibers, graphene derivatives, and carbon black, provide excellent platforms for surface modification, facilitating improved electron transfer and increased sensitivity during the detection of APAP and IBU analytes, due to their ability to detect trace analytes, minimize interference, and operate effectively in complex aqueous matrices.

Taouri and his colleagues., [81] have used a non-metallic composition, such as carbon black nanopowder, to modify the surface of a cavity microelectrode, resulting in the sensor referred to as nCB/CME. This sensor was then used to measure APAP concentrations in drinking water and in wastewater samples collected from the wastewater treatment plant in Betton (Rennes Metropole, France). The structure of carbon black nanoparticles, characterized by high porosity, large surface area, and high conductivity [82,83], makes them highly suitable for electrochemical applications. Surface modification of electrodes with nCB significantly enhances analytical performance, including improvements in detection limits, sensitivity, reproducibility, and repeatability [84]. Moreover, due to its low cost [85], carbon black can serve as a platform for the development of other nanomaterials [83,86]. Square wave voltammetry (SWV) combined with an electrochemical preconcentration (EP) technique at 0 V for 900 s, was employed to detect APAP on nCB/CME within the potential range of 0 to 0.8 V in Britton Robinson solution (BRS) 0.04 M, pH 7, which included acidic components (H_3PO_4, CH_3COOH, and H_3BO_3 at 0.04 M) and an alkaline component (NaOH 0.02 M). Two linear ranges were registered, from 5×10^{-10} to 9×10^{-8} mol/L and from 1×10^{-7} to 1×10^{-6} mol/L. The limit of detection (LOD) achieved using the EP method was 1.40×10^{-10} M, and the limit of quantification (LOQ) was 4.30×10^{-10} M. To validate the method, the standard addition approach was applied, as pre-spike concentrations were below LOD. The sensor exhibited a recovery range from 101 % to 104 %, and the RSD values were below 5 % for different samples, demonstrating its great potential for real environmental monitoring applications.

Traces of paracetamol have also been detected in surface waters [87], including rivers. To confirm the presence of this compound in these sources, Yi and his team [88] analysed the Fenhe River in Shanxi Province, China, using a modified glassy carbon electrode based on ultrathin graphitic carbon nitride nanosheets doped with graphene oxide (g-CNNS/GO/GCE). Unmodified graphitic carbon nitride (C_3N_4) has limitations in electrochemical applications due to its low conductivity and large particle size [89,90]. To enhance its electrochemical performance, graphitic C_3N_4 is exfoliated into CNNS, which improves its conductivity [91]. Further modification with two-dimensional (2D) materials such as graphene oxide (GO) provides additional benefits for environmental electrochemical measurements, including a wide linear range, high repeatability and reproducibility, and effective interference suppression [92–94]. Differential pulse voltammetry (DPV) was used to detect APAP in 0.1 M phosphate-buffered-saline (PBS) at pH 5. The linear concentration ranges were 5×10^{-7} to 3×10^{-5} mol/L. Experimental measurements confirmed a detection limit of 1.7×10^{-8} mol/L for river water samples. Recovery ranged from 95.6 % to 111.8 %, with a RSD of 4.53 %.

The concentration of APAP in river and seawater was also studied by Sasal [95] using screen-printed carbon electrodes modified with carbon nanofibers (SPCE/CNFs). Differential pulse adsorptive stripping voltammetry (DPAdSV) was employed with an accumulation time of 90 s at 0 V to analyze samples from the Bystrzyca River, the Vistula River, and the Baltic Sea. Because the initial APAP concentrations in real samples were below the LOD (5.4×10^{-10} mol/L), the standard addition method was applied. Sulfuric acid (0.1 M) was used as the supporting electrolyte. The sensor showed two linear ranges: from 2×10^{-9} to 5×10^{-8} mol/L and from 1×10^{-7} to 2×10^{-6} mol/L. Recovery ranged from 96.2 % to 104.6 %, with RSD values between 1.8 % and 5.1 %.

In the study by Tseng et al., [96] the use of metal phosphates, specifically strontium phosphate (SrP), was investigated. SrP was cross-linked with g-CNNS and used to modify the surface of SPCE for the detection of APAP in river and pond water samples. Metal phosphates

exhibit excellent electrical, thermal conductivity and chemical stability [97]. Their electrocatalytic performance is significantly enhanced when integrated into nanocomposites [98]. Linear sweep voltammetry (LSV) was performed in 0.05 M PBS at pH 7. The sensor demonstrated a linear detection range from 1.0×10^{-8} to 3.7×10^{-4} mol/L and a LOD of 0.002 μM. Recovery values were 95.8 % to 98.9 % in river water and 99.2 % to 99.5 % in pond water.

Serrano et al., [99] investigated the presence of APAP and IBU in hospital wastewater and tap water in Tunisia. They developed and evaluated sensors using screen-printed electrodes modified with various carbon-based nanomaterials, including multi-walled carbon nanotubes (SPCNTE), carbon nanofibers (SPCNFE), and graphene (SPGPHE). Among them, SPCNFE showed the highest sensitivity and was used for real sample analysis. The linear detection range for APAP was 5.95×10^{-7} to 5.29×10^{-6} with an LOD of 0.03 mg/L, while for IBU it was 2.2 – 10.2 mg/L with an LOD of 0.6 mg/L, using DPV method in 0.1 M acetate buffer at pH 5.5. The standard addition method applied to tap water yielded recoveries of 103.1 % (APAP, RSD 8.93 %) and 99.5 % (IBU, RSD 0.96 %). In unspiked hospital wastewater, APAP was detected at 1.17 mg/L, while IBU was undetectable.

In our opinion, nCB nanopowder sensor offers the best sensitivity with the lowest detection limit, although its long preconcentration time limits rapid analysis. The metal phosphate sensor, combined with g-CNNS, provides a wider linear range and faster detection but still requires further validation in complex environmental samples.

IBU has been detected in surface water sources [100]. For this purpose, Motoc et al., [101] developed two voltametric sensors using different allotropic forms of nanocarbon. One sensor used a fullerene–carbon nanofiber paste electrode (FULL/CNF) for DPV detection, and the other used a graphene–multi-walled carbon nanotube paste electrode (GR/MWCNT) for detection via multiple-pulsed amperometry (MPA). Allotropic forms of carbon are widely employed due to their high sensitivity, fast response, excellent selectivity, and adsorptive capabilities, stemming from their large surface area and porosity [102–104] . Water samples were collected from the Bega River, Romania. The FULL/CNF sensor achieved a LOD of 0.18 nM, LOQ of 0.54 nM, with RSD of 0.05 % while the GR/MWCNT had a LOD of 0.64 nM, LOQ of 1.92 nM and RSD of 0.06 %, using 0.1 M Na_2SO_4 as the supporting electrolyte. Both electrodes showed strong reproducibility and stability and were capable of simultaneous IBU detection in aqueous solutions. Recovery values for both sensors ranged from 97.60 to 99.10 %. We consider that the FULL/CNF sensor demonstrates higher sensitivity and precision than the GR/MWCNT sensor, making it more suitable for detecting low concentrations of ibuprofen in water.

4. Electrochemical sensors fabricated by metallic nanoparticles

Metal nanoparticles (MNPs) have exceptional physicochemical properties, including high surface-to-volume ratio, excellent electrical conductivity, and strong catalytic activity [105]. The size, shape, and surface chemistry of MNPs further allow the design of sensors tailored for specific environmental targets, including APAP and IBU, making them ideal candidates for real-world water monitoring applications.

APAP has also been detected in groundwater, and Weheabby et al., [106] developed a sensor by modifying the surface of a screen-printed carbon electrode (SPCE) with silver nanoparticles (AgNPs) and carboxylated multi-walled carbon nanotubes (COOH-MWCNT) (*see* Fig 5). According to the authors, the proposed AgNPs/COOH-MWCNT/SPCE sensor exhibits good sensitivity and selectivity for the determination of APAP in water samples. When these two nanomaterials are combined, composite is formed from these two nanomaterials, their individual properties are integrated, enhancing the overall performance of the sensor [107–109]. MWCNTs are known for their high stability and exhibit excellent adsorptive and electrocatalytic properties [110,111]. Multiple studies have demonstrated that functionalizing MWCNTs with carboxyl (COOH) groups significantly improves surface properties, enhances substrate adhesion, boosts electrochemical performance, and increases surface reactivity for further functionalization steps [112–114]. Meanwhile, AgNPs are widely used for surface modification due to their favourable electronic, sensing, and electrocatalytic properties [115,116]. Groundwater samples for this study were collected by the Saxony State Office for Environment, Germany. Electrochemical measurements were performed using the electrochemical method SWV, coupled with the standard addition method as the analytical technique, under optimal conditions (pH 7.4, 0.1 M, PBS). The sensor exhibited recovery values ranging from 98.1 % to 99.2 %, with RSD values ranging between 0.67 % and 2.65 % The linear detection range extended from 0.5 to 1000 μM, with a LOD value of 0.24 μM.

Haššo et al., [117] investigated the concentration of APAP in tap water using SPCE modified with gold nanoparticles (AuNPs). The novelty of this study lies in the comparison of two measurement approaches: one performed with an automated benchtop electrochemical station (BES), and the other manually operated by a laboratory technician. Both approaches were conducted under identical experimental conditions. SPCEs are known for their ease of modification [118,119], and efforts are ongoing to automate their use. The BES system enables full automation for SPCE testing. The electrochemical method employed was DPV in Britton-Robinson buffer (BRS) at pH 4. The developed AuNPs/SPCE sensor demonstrated a linear detection range of 5.0×10^{-6} to 2.0×10^{-4} mol/L, with a LOD of 3.8 μM at a deposition potential (E_{DEP}) of −0.1 V. The manually performed measurements yielded a recovery rate of 95.7 % with a RSD of 7 %, while the automated approach achieved a higher recovery of 103.4 % and a lower RSD of 3 %.

The study by Alkhudaydi et al., [120] explored the detection of APAP in seawater using a nanosensor based on an MXene material such as Ti_3C_2, which is combined with iron oxide nanoparticles (Fe_3O_4) used to modify the surface of a GCE (Fe_3O_4/Ti_3C_2/GCE). MXenes are2D materials composed of transition metal carbides, nitrides, or carbonitrides [121]. Ti_3C_2 has shown excellent electrocatalytic properties for APAP oxidation, enablingsimultaneous detection and identification [122]. Its unique features, such as a layered structure, tunable optical bandgap, large surface area, and chemical diversity, make it ideal for sensing applications [123,124], as well as in photocatalysis [125], energy storage [126], and chemical sensors [127]. Iron oxide nanoparticles further enhance the sensor's performance due to their magnetic properties, biocompatibility, and high surface area [128–130]. Using DPV, the sensor achieved a very low LOD of 0.63 nM with an RSD between 3.07 % and 3.42 %.

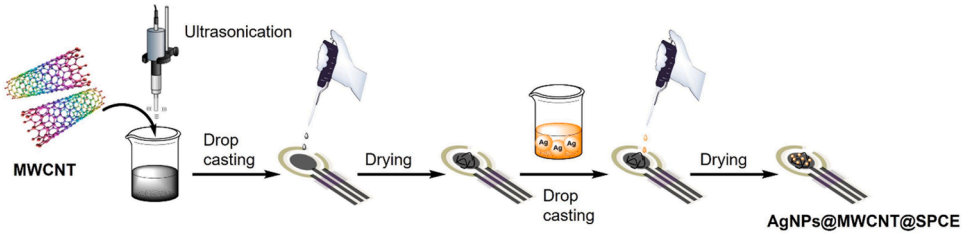


Fig. 5. Schematic of APAP detection based on AgNPs/COOH-MWCNT. Reprinted with permission from ref [106].

Akhtar et al., [131] developed a green-synthesized nanocomposite of titanium oxide–aluminum oxide ($TiO_2 \times Al_2O_3$ NC) to modify a gold electrode (AuE) for APAP detection. The nanocomposite was synthesized using an extract from the Fagonia plant via a co-precipitation method. Green synthesis is increasingly adopted for its environmental benefits [132], with plant extracts serving as both reducing and stabilizing agents [133,134]. The sensor, tested using DPV in 0.1 M PBS at pH 7, was applied to hospital wastewater and river water samples. It exhibited a LOD of 0.017 μM and a linear range of 9×10^{-8} to 6.5×10^{-5} mol/L. Recovery values ranged from 96.26 % to 104.51 % in hospital wastewater (RSD: 2.12–3.74 %) and from 98.2 % to 102.4 % in river water (RSD: 1.81–3.02 %).

Mohiuddin et al., [135] investigated APAP detection in bottled drinking water using a ternary nanocomposite sensor composed of palladium-doped-α-MnO_2 nanorods supported on graphene (Pd@α-MnO_2/G), applied on a GCE. The hollandite-type structure of a α-MnO_2 offers multiple doping sites due to its octahedral cavities [136,137]. Palladium, as a noble metal, is known to enhance surface electrochemical properties significantly [138]. The sensor was tested using DPV in 0.1 M PBS at pH 7.4, exhibiting a linear range of 1.0×10^{-7} to 3.75×10^{-4} mol/L, a LOD of 0.059 μM, and recovery between 97.8 % and 101.3 %, with an RSD of 1.8 %.

Kang et al., [139] developed an eco-friendly sensor using carbon derived from barley leaves, activated with zinc chloride ($ZnCl_2$), and decorated with AuNPs for APAP detection in tap and lake water. Biomass-derived carbon offers a high surface area and porosity, enhancing adsorption and reaction efficiency [140,141]. $ZnCl_2$ serves as an effective activator by increasing porosity and surface area [142,143]. Using DPV in BRS (pH 4), the sensor exhibited two linear ranges: 1.0×10^{-7} to 1.5×10^{-5} mol/L and 1.5×10^{-5} to 3.0×10^{-4} mol/L, with a LOD of 0.02 μM. Recovery ranged from 95.5 % to 103.6 % in tap water and from 93.7 % to 107.7 % in lake water, with RSD values below 5 % in both cases.

Our evaluation indicates that the AgNPs/COOH-MWCNT-modified SPCE provides a reliable combination of sensitivity and selectivity with low RSDs, while the MXene-based sensor offers the best detection limit and sensitivity. Despite these advances, challenges remain in achieving consistent detection at ultra-trace levels.

Mokaba et al., [144] focused on the detection of IBU in wastewater using SPCEs modified with manganese oxide nanoparticles (MnO_2NPs/SPCE). Samples were collected from the Northern Wastewater Treatment Plant in Gauteng Province, South Africa. PBS at pH 7 was selected as the supporting electrolyte, as the oxidation peak was found to be pH sensitive. Peak currents increased from pH 5 to 7 but decreased beyond pH 7. DPV was used as an electrochemical technique. The sensor exhibited a linear detection range of 1×10^{-9} to 6×10^{-8} mol/L, with an LOD of 0.4 mM. Recovery values ranged from 105 % to 110 %, with an RSD of 0.151 %, highlighting the sensor's strong potential for wastewater monitoring. The sensor shows strong sensitivity, though its performance is influenced by pH.

5. Electrochemical sensors fabricated from metal organic frameworks

Metal-organic frameworks (MOFs) are highly organized crystalline structures created by assembling metal ions or clusters with organic linking molecules [145] and they have gained significant attention in the field of sensor development [146] due to their highly porous structures, chemical properties, and large surface areas [147,148]. Their capacity to selectively capture and interact with specific target molecules makes them excellent materials for detecting APAP and IBU with high sensitivity and precision [149,150]. In electrochemical sensors, MOFs can improve conductivity by incorporating conductive metal centers or by forming composites with nanomaterials like graphene or metal nanoparticles [151].

Another sensor for the determination of APAP in tap water was developed by Theyagarajan et al. [152] using cobalt hexacyanoferrate (CHCF) incorporated into an iron terephthalate metal organic framework (TMF). GCE was used as the working electrode. Following the standard addition technique, known concentrations of APAP were added to tap water, and the analyte concentration was determined using DPV in a 0.1 M KCl electrolyte. The experimental results indicated that the LOD for the CHCF/TMF/GCE sensor was 13 μM, with a linear detection range from 5.0×10^{-5} to 7.37×10^{-3} mol/L and RSD values ranged from 3.7 to 4.2 %, and the recovery was in the range 96.2 to 106.2 %.

Yang et al., [153] developed a sensor for APAP detection in drinking water and samples from the Chenjiang River in China. The working electrode was GCE modified with cobalt single-atom anchored porous carbon composites doped with nitrogen (Co–N–C@PC). The Co–N–C@PC composite was prepared via one-step pyrolysis using a Co/Zn zeolitic imidazolate framework as the precursor. Electrochemical measurements were performed using DPV in 0.1 M PBS at pH 7. The sensor showed a linear detection range of 1.0×10^{-7} to 2.5×10^{-5} mol/L and a LOD of 34.1 nM. Recovery in tap water ranged from 89.4 % to 102.0 % (RSD: 2.12–3.92 %), while for river water, it ranged from 91.1 % to 95.0 % (RSD: 1.14–3.90 %).

Guan et al., [154] created a sensor from polyoxometalate based metal organic framework (POMOF) for detecting APAP in pharmaceutical industrial wastewater and water from Yulong Snow Mountain. Phosphotungstate (PW_{12}) was encapsulated in a Cu-based MOF (Cu-BTC) via a solvothermal method to form Cu-BTC@PW_{12}, which was immobilized on GO to enhance conductivity and electron transfers [155]. The Cu-BTC@PW_{12}/GO composite was drop-cast onto a GCE and tested via DPV in 0.1 M PBS (pH 6), using the standard addition method. The recovery in wastewater ranged from 100.1 % to 108.0 % (RSD: 0.4–1.1 %), and for natural water, from 98.3 % to 100.5 % (RSD: 0.7–1.4 %). The LOD was 62.3 nM.

Ji and his team [156] developed a 2D MOFs electrochemical sensor for determining the amount of APAP in river water. To construct the sensor, they used a MOF material composed of a series of nanosheets of copper tetrakis(4-carboxyphenyl) porphyrin (Cu-TCPP), which were prepared by ultrasonic exfoliation using different organic solvents, (*see* Fig 6). This material was incorporated into a GCE. The Cu^{2+} ion as a coordinated metal center can undergo redox reactions, passing into reduced transition states such as Cu^{+} or Cu^{0}, or into oxidized transition states such as Cu^{3+} and Cu^{4+}, making this MOF electroactive [157]. In the preparation of nanosheets, the choice of solvent plays a crucial role [158,159]. A suitable solvent, such as DMF in this case, increases the yield of exfoliated nanosheets and stabilizes them [160]. As a result, the sensor created with these materials exhibits high redox activity, faster electron transfer rate, and a larger electrochemical reaction area. The measurements of real samples were performed using the standard addition method in a 0.1 M PBS with pH 7 by employing the DPV technique. The Cu-TCPP-DMF/GCE sensor demonstrated a LOD of 1.9 nM. The standard addition method resulted in a recovery range of 103.4 to 104.2 % and an RSD of 4.2 %.

Dong et al., [161] fabricated a sensor based on a bimetallic NiCo-MOF structure grown on carbon cloth (CC) using a solvothermal method, followed by a selenization step to form NiCoSe/CC electrodes for the detection of APAP in tap water and in the Songhua River (China). These bimetallic selenides benefit from tunable valence states and electronic configurations, improving electrocatalytic efficiency [162,163]. To evaluate the performance of the NiCoSe/CC sensor, measurements were conducted in a 0.2 M PBS with pH 6, using DPV method. The sensor exhibited a linear range of 5.0×10^{-8} to 2.0×10^{-4} mol/L and a LOD of 0.012 μM. The recovery range in tap water samples was 95.5 to 102.4 % with an RSD of 2.9 to 3.8 %, while in river water, the recovery was 98 to 104.8 % with an RSD of 3.7 to 4.2 %.

Xie and his team [164]developed a 3D hybrid material of $ZnCo_2O_4$@Fe_3O_4/MWCNTs using MOF compounds ZIF-8/67@Fe_3O_4 as a precursor for the determination of APAP in lake water. The core–shell $ZnCo_2O_4$@Fe_3O_4 was obtained through pyrolysis using

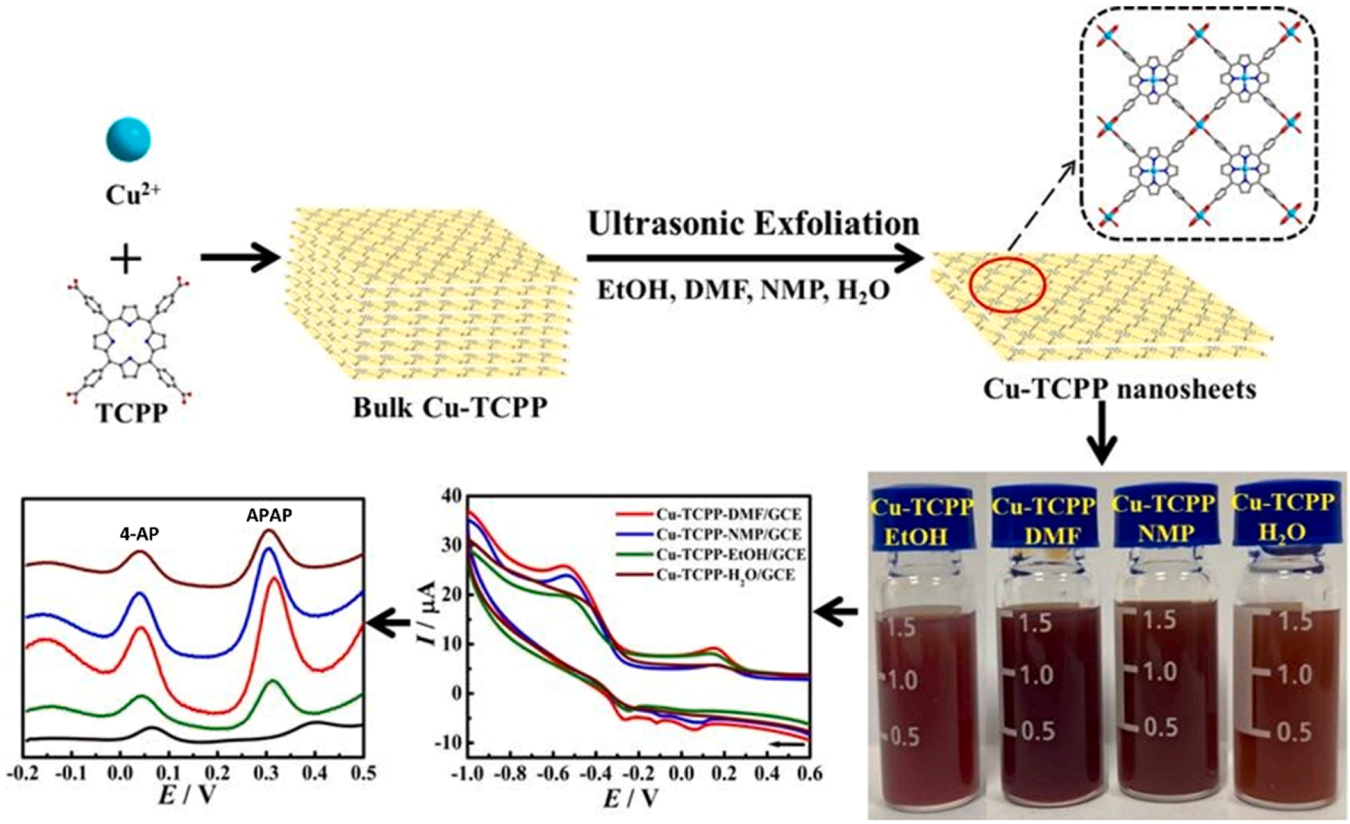


Fig. 6. Schematic diagram of APAP detection based on Cu-TCPP/GCE. The graph shows the simultaneous determination of APAP and 4-aminophenol. Reprinted with permission from ref [156].

ZIF-8/67@Fe_3O_4 as a precursor material. To enhance electrical conductivity, a conductive MWCNTs substrate was incorporated, resulting in the formation of 3D nanocomposites. This 3D composite was used to modify the GCE. The sensor showed a linearity range from 1.0×10^{-6} to 1.5×10^{-4} mol/L and a LOD of 0.018 μM and the measurements were performed in PBS 0.2 M with pH 7 using DPV method. From the real samples analysis, the interval of recovery ranged from 97.70 to 98.48 % and the RSD value 2.57 to 3.20 %.

In another study, Dong et al., [165] developed a 3D multiporous nanorod structure composed of Co- and N-co-doped molybdenum dioxide/molybdenum carbide (MoO_2/MoC) to determine APAP in river water. The nanorod hybrids were effectively prepared by heating MoO_3 nanorods coated with a ZIF-67 core-shell structure, followed by a carburization process at elevated temperatures. To create these porous materials derived from transition metals, MOFs from the ZIF-67 group are often used due to their high thermal stability and rich content of C, Co, and N [166,167]. Using ZIF-67 as a carbon precursor is a promising strategy to combine it with Mo-based materials, resulting in 3D porous MoO_2/MoC hybrids doped with N and Co. The 3D material was used to modify the surface of the GCE, and measurements conducted using the DPV method in a 0.2 M PBS at pH 6 resulted in a linear range of 0.05 to 200 μM, with a LOD of 5.0×10^{-8} to 2.0×10^{-4} mol/L. Recovery tests performed on water samples from the Xinkai River, Shenyang, showed a recovery range of 96 to 99.7 %.

Gu and his colleagues [168] constructed a zinc-based MOF (MOF-5) composite sensor for APAP detection by combining MOF-5 with MWCNTs via an in-situ solvothermal method. The composite was used to modify a GCE and tested using DPV in PBS at pH 7 for real tap water analysis. The sensor showed a linear range of 1.5×10^{-4} to 6.0×10^{-4} mol/L and an LOD of 0.061 μM. Recovery ranged from 98.2 % to 104.2 %, with an RSD below 5 %.

Evaluation of these MOF-based sensors shows that Cu-TCPP nanosheets and NiCoSe/CC provide excellent sensitivity and low detection limits, ideal for APAP detection in various waters.

6. Electrochemical sensors fabricated from synthetic polymers

Synthetic polymers are increasingly used in electrochemical sensors for the detection of APAP and IBU in environmental waters. Their ease of modification, and ability to form conductive matrices make them ideal for this purpose [169]. When combined with nanomaterials or carbon-based compouns, these polymers contribute to improved electron transfer and efficient analyte recognition, enabling accurate monitoring in complex real-world samples.

Pollap et al., [170] in their workmodified the surface of a graphite electrode (GE) with a composite of titanium dioxide sol enriched with mesoporous carbon CMK-3, gold nanoparticles and Nafion (TiO_2/AuNPs/CMK-3/Nafion/GE). CMK-3 possesses an ordered nanorod structure, uniform pore size, and a large specific surface area [171]. CMK-3 offers a high density of active sites for efficient electron and proton transfer, making it highly suitable for a wide range of applications in electrochemical sensors [172–174]. To evaluate the potential for sensor applications, the simultaneous determination of APAP was performed in environmental water samples from the Wisla and Raclawka rivers (Poland) using CV in PBS 0.1 M with pH 5. From LOD was determined to be 0.21 μM, with two linearity intervals: 1.0×10^{-6} to 1.0×10^{-5} mol/L and 1.0×10^{-5} to 5.2×10^{-5} mol/L. Measurements in the Vistula River, using the standard addition method, resulted in an RSD of 3.3 %, while in the Racławka River, the RSD was 2.5 %.

Sekar et al., [175] developed a sensor based on the synthetic polymer poly Azovan Blue (PAB) combined with graphitic carbon nitride (g-C_3N_4) to modify the surface of a screen-printed carbon electrode (SPCE) for APAP detection in tap water. Azure B, a polychromed derivative of methylene blue, undergoes oxidative demethylation and exhibits low

solubility and strong self-aggregation, which affects its electrochemical behavior and polymerization efficiency [176,177]. Measurements using the g-C_3N_4/PAB/SPCE sensor were performed in a 0.1 M PBS at pH 7, employing the DPV method. The sensor exhibited a LOD of 0.016 μM, with a linearity range of 8.0×10^{-8} to 7.5×10^{-5} mol/L. Recovery values ranged from 95.2 to 97.4 %, with an RSD of 1.37 to 2.15 %.

In another study performed by Yuan et al., [178] zeolitic imidazolate framework (ZIF-67) was utilized as a precursor to develop an electrochemical sensor for the detection of APAP in water samples collected from the Wujin District River in Changzhou, China. This MOF is composed of cobalt ions and 2-methylimidazole, and it is known as a versatile precursor for the production of advanced functional materials due to its extraordinary adaptability [179]. ZIF-67 was mixed with nickel and thermally treated, yielding Co_3O_4@$NiCo_2O_4$, a core-shell structured material with enhanced capabilities for catalysis and electrochemical sensing [180]. To further optimize the conductivity of the electrode material, a simple one-step carbonization process was employed to synthesize chitosan hydrogel-derived nitrogen-sulfur co-doped porous carbon (CHPC—CS) enhancing its electrochemical performance [181]. The measurements were performed using the DPV method with GCE as a working electrode in PBS 0.2 M with pH 7. The Co_3O_4@$NiCo_2O_4$/CHPC—CS/GCE sensor demonstrated a LOD value of 0.068 μM, a linear detection ranges from 5.0×10^{-7} to 4.7×10^{-4} mol/L, with recovery values between 99.10 and 104.24 %, and an RSD ranging from 0.98 to 1.20 %.

Guo et al., [182] used ZIF-67 as an active component (not precursor) and physically combined it via ultrasonic treatment with carboxylated MWCNTs and Nafion to modify a GCE. The resulting sensor, ZIF-67/MWCNT–COOH/Nafion/GCE (*see* Fig. 7), was used to detect APAP in Yellow River water. Nafion helped improve the signal-to-noise ratio in voltammetric detection [183,184]. Measurements using the developed sensor were performed in a 0.1 M PBS at pH 6.5, using DPV technique. The sensor exhibited a linear detection range of 1.0×10^{-7} to 2.0×10^{-5} and 2.0×10^{-5} to 1.0×10^{-4}. The LOD was determined to be 0.07 μM, with a recovery range of 94.7 to 99.6 % and RSD between 1.1 to 4.7 %

Shalauddin et al., [185] analyzed water from Tasek Varsiti Lake, Malaysia, using a sensor composed of nanocellulose and polypyrrole (NNC–PPY), drop-cast with single-walled carbon nanotubes (SWCNTs) on a screen-printed electrode (*see* Fig. 8). The growing need for eco--friendly detection methods has increased attention toward cellulose-derived nanomaterials [186], which are being explored for the development of advanced electrochemical sensors [187,188] because of peculiar characteristics such as high porosity, expansive surface area, strong adhesion, and exceptional biocompatibility [189]. To improve the conductive properties of nanocellulose, polypyrrole is one of the most used conductive polymers in the design of sensors [190,191]. The NNG-PPY/SWCNTs/SPE sensor demonstrated a LOD of 0.072 nM and a linearity range of 5.4×10^{-8} to 4.0×10^{-5} mol/L in 0.1 M PBS at pH 7, using the SWV technique. When applied to real sample measurements, the recovery range was observed to be between 98 and 105 %, with an RSD ranging from 0.75 to 2.35 %.

Based on this analysis, sensors combining advanced carbon materials with metal oxides or MOFs show strong sensitivity and good linear ranges for APAP detection.

Sharma et al., [192] developed a sensor using ferric oxide, chitosan, and grafted polypyrrole (α-Fe_2O_3-CHIT-g-PPy) to monitor the presence of residual IBU in the wastewater from hospitals in Sonipat, India.

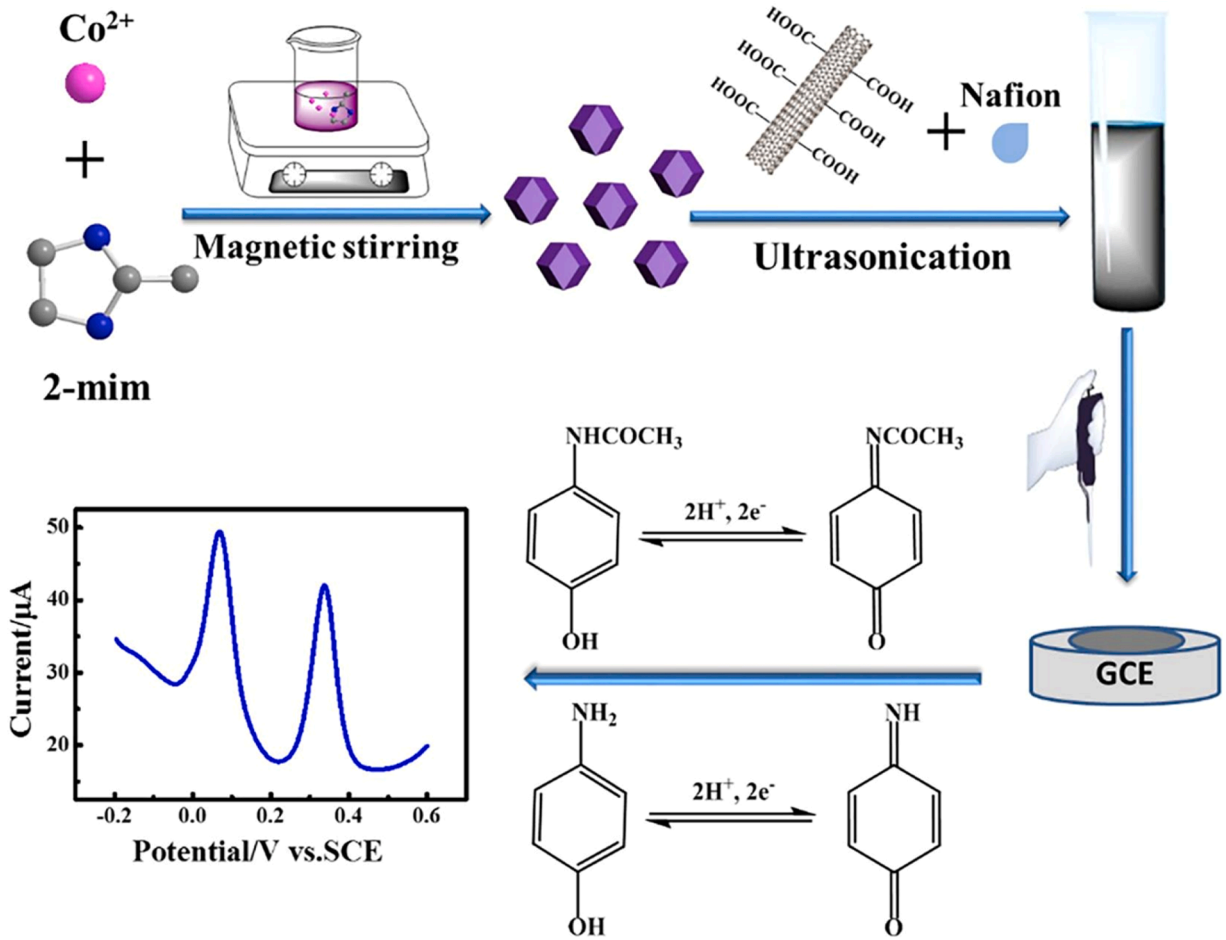


Fig. 7. Schematic diagram based on ZIF-67/MWCNT-COOH/Nafion/GCE sensor for APAP detection and 4-aminophenol. Reprinted with permission from ref [182].

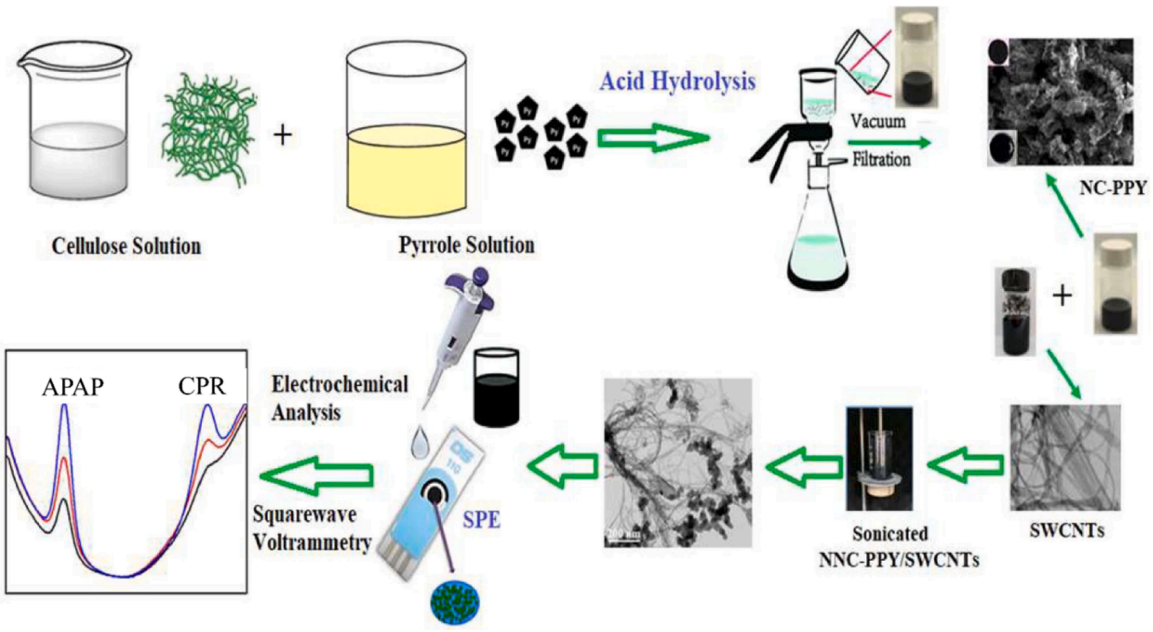


Fig. 8. Schematic diagram of NNC-PPY/SWCNTs/SPE, for simultaneous APAP and ciprofloxacin (CPR). Additionally, electron microscopy images are presented for each step of the sensor fabrication process. Reprinted with permission from ref [185].

Chitosan, a biopolymer sourced from chitin, has garnered significant attention in the field of electrochemical sensor development [193]. Its desirable characteristics, including biocompatibility, biodegradability, and a high surface area, make it a valuable material for such applications [194,195]. Additionally, CHIT is highly adaptable and can be readily functionalized, enhancing its suitability for creating advanced electrochemical sensors [196,197]. The α-Fe_2O_3/CHIT/g-PPy sensor Fig 9, integrated into a film-coated glass plate (Indium Thin Oxide electrode), exhibited a linear range of 0.5×10^{-6} to 1.0×10^{-4} mol/L, measured using the potentiometric method in a buffer solution with pH 7. Real measurements from the hospital revealed wastewater a recovery between 98.0 to 102.2 %, while for the other hospital, the recovery ranged from 98.4 to 101.4 %. From our perspective, the sensor offers good practical applicability due to its strong recovery values, though the use of potentiometry may limit its sensitivity compared to voltammetric techniques.

Table 2 summarizes the performance of electrochemical sensors modified with various materials for the detection of APAP and IBU in different water samples. These sensors engineered to improve sensitivity, specificity, and overall detection performance. The table includes key parameters such as the electrochemical techniques employed, LOD, linear detection ranges, supporting electrolytes, recovery ranges, and RSD.

To conclude our study, we suggest that whenever a new electrochemical sensor is developed, the possibility of its recyclability should be considered. This approach would offer significant benefits in terms of the overall sustainability of the sensor's life cycle. Usually, the surface of GCEs can be cleaned with alumina slurry and modified again. This process can be demanding but provides reproducible results. Colin et al., suggest oxygen plasma treatment as a more effective method for cleaning and activating GCEs [198]. Previously, SPEs were typically used as single-use sensors. However, the increasing demand for these electrodes in laboratory applications has led to the development of cleaning and regeneration methods, allowing for their reuse [199]. Chemical cleaning using piranha solution (a mix of sulfuric acid and hydrogen peroxide in a ratio of 3:1) [200] is the most used method.

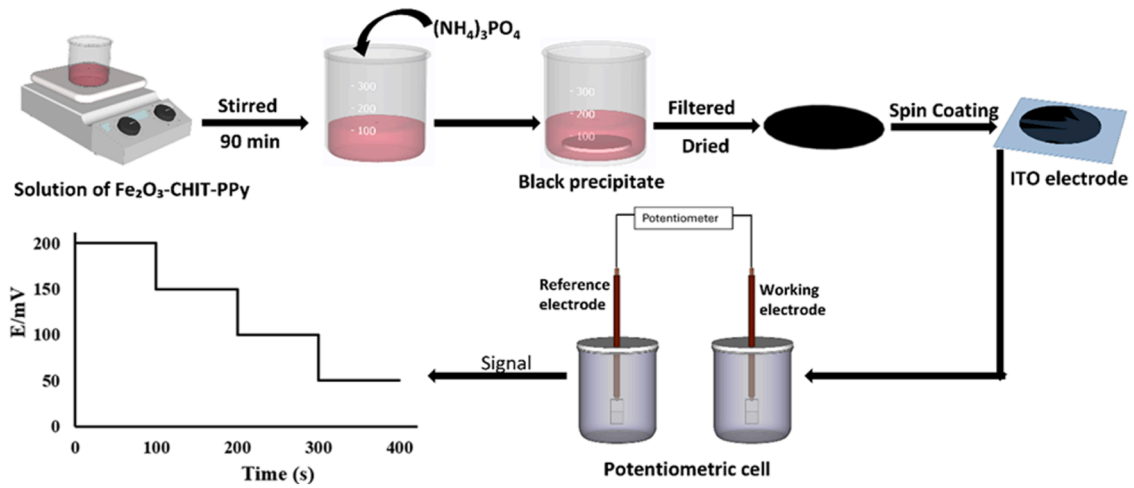


Fig. 9. Schematic diagram of α-Fe_2O_3/CHIT/g-PPy sensor.

Table 2
Overview of sensors used for the determination of APAP and IBU in water samples.

Compound	*Sensor matrix*	*Electrochemical method*	*Analytical Parameters*	*Supporting electrolyte*	*RSD (%)*	*Recovery range (%)*	*Sample*	*Ref*
APAP	nCB/CME	SWV	Linear range (M): 5×10^{-10} to 9×10^{-8} 1×10^{-7} to 1×10^{-6} LOD (μM): 0.00014 LOQ (μM): 0.00043	BRS 0.04 M pH=7	<5	101 - 104	drinking water; wastewater	[81]
	g-CNNs/GO/GCE	DPV	Linear range (M): 5×10^{-7} to 3×10^{-5} LOD (μM): 0.017	PBS 0.1 M pH=5	4.53	95.6 - 111.8	river	[88]
	SPCE/CNFs	DPAdSV	Linear range (M): 2×10^{-9} to 5×10^{-8} 1×10^{-7} to 2×10^{-6} LOD (μM): 0.00054	Sulfuric acid 0.1 M	1.8–5.1	96.2 - 104.6	river; sea	[95]
	SrP/g-CNNSs/SPCE	LSV	Linear range (M): 1.0×10^{-8} to 3.7×10^{-4} LOD (μM): 0.002	PBS 0.05 M pH=7	–	95.8 - 98.9 99.2 – 99.5	river; pond	[96]
	SPCNFE	DPV	Linear range (M): 5.95×10^{-7} to 5.29×10^{-6} LOD (μM): 0.198	Acetate buffer solution 0.1 M pH=5.5	8.93	103.1	wastewater (con, 1.17 mg/L); tap water	[99]
	AgNPs/HOOCMWCNT/SPCE	SWV	Linear range (M): 5×10^{-7} to 1×10^{-3} LOD (μM): 0.24	PBS 0.1 M pH=7.4	0.67–2.65	98.1–99.2	groundwater	[106]
	AuNPs/SPCE	DPV	Linear range (M): 5.0×10^{-6} to 2.0×10^{-4} LOD (μM): 3.8	BRS pH=4	7 3	95.7 103.4	tap water	[117]
	Fe_3O_4/Ti_3C_2/GCE	DPV	LOD (μM): 0.00063	–	3.07 – 3.42	–	sea water	[120]
	TiO_2 x Al_2O_3–NC/AuE	DPV	Linear range (M): 9.1×10^{-8} to 6.5×10^{-5} LOD (μM): 0.017	PBS 0.1 M pH=7	2.12 – 3.74 1.81 – 3.02	96.26 – 104.51 98.2 – 102.4	wastewater; river	[131]
	Pd/α-MnO_2/G/GCE	DPV	Linear range (M): 1.0×10^{-7} to 3.75×10^{-4} LOD (μM): 0.059	PBS 0.1 M pH=7.4	1.8	97.8 – 101.3	bottled drinking water	[135]
	AuNPs/ZnAC/GCE	DPV	Linear range (M): 1.0×10^{-7} to 1.5×10^{-5} 1.5×10^{-5} to 3.0×10^{-4} LOD (μM): 0.02	BRS pH=4	<5	95.5 – 193.6 93.7–107.7	tap water; lake	[139]
	CHCF/TMF/GCE	DPV	Linear range (M): 5.0×10^{-5} to 7.37×10^{-3} LOD (μM): 13	0.1 M KCl	3.7 – 4.2	96.2 – 106.2	tap water	[152]
	Co-N-C@PC/GCE	DPV	Linear range (M): 1.0×10^{-7} to 2.5×10^{-5} LOD (μM): 0.0341	PBS 0.1 M pH=7	2.12 – 3.92 1.14 – 3.9	89.4 – 102 91.1 – 95.0	tap water; river	[153]
	Cu-BTC@PW12/GO/GCE	DPV	LOD (μM): 0.0623	PBS 0.1 M pH=6	0.4 – 1.1 0.7 – 1.4	100.1 – 108 98.3 – 100.5	wastewater; surface water	[154]
	Cu-TCPP-DMF/GCE	DPV	LOD (μM): 0.0019	PBS 0.1 M pH=7	4.2	103.4 – 104.2	river	[156]
	NiCoSe/CC	DPV	Linear range (M): 5.0×10^{-8} to 2.0×10^{-4} LOD (μM): 0.012	PBS 0.2 M pH=6	2.9 – 3.8 3.7 – 4.2	95.5 – 102.4 98.0 – 104.8	tap water; river	[161]
	$ZnCo_2O_4$@Fe_3O_4/MWCNTs/GCE	DPV	Linear range (M): 1.0×10^{-6} to 1.5×10^{-4} LOD (μM): 0.018	PBS 0.2 M pH=7	2.57 – 3.20	97.7 – 98.48	lake	[164]
	Co,N-MoO_2/MoC/GCE	DPV	Linear range (M): 5.0×10^{-8} to 2.0×10^{-4} LOD (μM): 0.013	PBS 0.2 M pH=6	–	96.0 – 99.7	river	[165]

(*continued on next page*)

Table 2 (*continued*)

Compound	*Sensor matrix*	*Electrochemical method*	*Analytical Parameters*	*Supporting electrolyte*	*RSD (%)*	*Recovery range (%)*	*Sample*	*Ref*
	MOF-5/MWC—NTs/GCE	DPV	Linear range (M): 1.5×10^{-4} to 6×10^{-4} LOD (μM): 0.061	PBS 0.1 M pH=7	<5	98.2 – 104.2	tap water	[168]
	TiO_2/AuNPs/CMK-3/ Nafion/GE	CV	Linear range (M): 1.0×10^{-6} to 1.0×10^{-5} 1.0×10^{-5} to 5.2×10^{-5} LOD (μM): 0.21	PBS 0.1 M pH=5	3.3 2.5	–	river	[170]
	g-C_3N_4/PAB/SPCE	DPV	Linear range (M): 8.0×10^{-8} to 7.5×10^{-5} LOD (μM): 0.016	PBS 0.1 M pH=7	1.37 – 2.15	95.2 – 97.4	tap water	[175]
	Co_3O_4@$NiCo_2O_4$/ CHPC—CS/GCE	DPV	Linear range (M): 5.0×10^{-7} to 4.7×10^{-4} LOD (μM): 0.068	PBS 0.2 M pH=7	0.98 – 1.2	99.1 – 104.24	river	[178]
	ZIF-67/MWCNT-COOH/ Nafion/GCE	DPV	Linear range (M): 1.0×10^{-7} to 2.0×10^{-5} 2.0×10^{-5} to 1.0×10^{-4} LOD (μM): 0.074	PBS 0.1 M pH=6.5	1.1 – 4.7	94.7 – 99.6	river	[182]
	NNG-PPY/SWCNTs/SPE	SWV	Linear range (M): 5.4×10^{-8} to 4.0×10^{-5} LOD (μM): 0.072	PBS 0.1 M pH=7	0.75 – 2.35	98.0 – 105.0	lake	[185]
IBU	SPCNFE	DPV	Linear range (M): 1.07×10^{-5} to 4.94×10^{-5} LOD (μM): 2.91	Acetate buffer solution 0.1 M pH=5.5	0.96	99.5	tap water	[99]
	FULL/CNF GR/MWCNT	DPV MPA	LOD (μM): 0.00018 LOQ (μM): 0.00054 LOD (μM): 0.00064 LOQ (μM): 0.00192	0.1 M Na_2SO_4	0.05 0.06	97.60–99.10 % for both sensors	river	[101]
	MnO_2NPs/SPCE	DPV	Linear range (M): 1×10^{-8} to 6×10^{-9} LOD (μM): 0.0004	PBS 0.1 M pH=7	0.151	105–110	waste water	[144]
	α-Fe_2O_3/CHIT/g-PPy	Potentiometric	Linear range (M): 0.5×10^{-6} to 1.0×10^{-4}	Buffer solution pH=7	–	98.0 – 102.2 98.4 – 101.4	wastewater	[192]

* All linear range units have been converted to M and all LOD units have been converted to μM.

7. Future research perspectives

Given the increasing global concern for environmental protection, many research groups have focused on developing various electrochemical sensors for monitoring pharmaceutical compounds in the environment. Despite promising advances, several critical gaps remain in sensor selectivity, real-time monitoring in complex matrices, and environmentally sustainable fabrication techniques. A key research direction involves the fabrication of sensors incorporating eco-friendly modifiers, such as green nanocomposites and natural polymers derived via sustainable processes. However, in our opinion, there is a lack of comparative studies in this field to compare such green sensors with conventional systems, making it difficult to evaluate the true environmental and analytical benefits.

Over the past five years, few biosensor studies have been reported for the detection of APAP and IBU in environmental water samples. One notable exception is the study by Herrera-Domínguez et al. [201], which demonstrated the detection of APAP in groundwater using bioelectrodes based on laccase enzymes immobilized on carbon paper electrodes modified with nanostructured molybdenum disulfide (MoS_2). In a green approach, the enzyme LacII was extracted from the native Mexican fungus *Pycnoporus sanguineus* CS43,and compared to a commercial laccase from *Trametes versicolor* (TvL). The natural LacII enzyme achieved a LOD of 0.5 μM, significantly outperforming the commercial enzyme, which had a LOD of 24.88 μM. From our research, local natural materials are often ignored in regular sensor design, even though they can perform well and cost less. For IBU, enzyme-based sensors remain largely unexplored, possibly due to the compound's more challenging electrochemical oxidation behavior. Further explorations are needed to increase the stability of enzyme-based biosensors for environmental monitoring. Bounegru et al. [202] have suggested that some biosensor limitations, such as interference from organic matter or fluctuating environmental conditions, could be addressed using aptamer-based biosensors. Aptamers offer high target specificity and maintain stability in complex matrices. In our opinion, while aptamers show great promise, their use in durable, field-ready sensors is still limited, mainly due to challenges such as degradation under environmental stress, which we find have not yet been fully addressed in the current literature. An example is the aptasensor proposed by Roushani et al. [203], which utilized a GCE modified with a nanocomposite of multi-walled carbon nanotubes, an ionic liquid, and chitosan (MWCNTs/IL/CHIT). An IBU-specific aptamer was covalently immobilized on the surface, and methylene blue (MB) was intercalated as a redox probe. The resulting sensor demonstrated excellent sensitivity, detecting IBU in spiked wastewater with a LOD as low as 20 pM. The use of MOFs is still in its early stages for the detection of IBUs, and systematic studies are required to evaluate long-term performance in natural waters.

Another promising direction is the application of MOFs as modifiers

for the electrochemical detection of IBU. A study by Motoc et al., [204] has demonstrated that IBU can be successfully detected using a composite electrode made from the HKUST-1 metal-organic framework and carbon nanofibers (HKUST-CNF) in spiked water samples. Real sample measures revealed that the sensor achieved an impressive recovery rate of 99 % with an RSD of 2.8 %, highlighting the effectiveness of this approach.

In addition to material innovation, the automation of the measurement process is a vital area that warrants greater attention. Automation can significantly increase productivity, eliminate the need for manual work, and reduce analysis time. Recent research shows that several prototype automated electrochemical systems have already been developed by different research groups. Examples include the benchtop electrochemical station (BES) developed by Zitka et al. [205], the electrolab by Oh et al. [206], eLab by A. Pence et al. [207], and the scanning droplet cell (SDC) developed by Dieckhöfer et al. [208].

In summary, future research in electrochemical sensing APAP and IBU should prioritize the development of green, sustainable sensor materials, the integration of automated and miniaturized systems, and the expansion of biosensing strategies. It is important to note that we believe that more standardized protocols for evaluating environmental sensors will be essential to enable real-world adoption. Addressing these priorities will accelerate the transition from laboratory prototypes to real-world environmental monitoring systems.

8. Conclusions

This review provides a comprehensive overview of relevant literature from the past five years on the electrochemical detection of APAP and IBU in various water matrices. The findings underscore the critical role of electrode surface modification in enhancing sensor performance, particularly in improving sensitivity, selectivity, and lowering detection limits to enable trace-level quantification. A wide array of materials, including metal-based nanostructures, carbon derivatives, MOFs, and synthetic polymers, have been employed with notable success in this regard.

Despite these advances, several avenues remain open for further exploration. Special emphasis should be placed on the integration of biosensing strategies, particularly enzyme- and aptamer-based systems, which offer high specificity and the potential for operation in complex matrices. In our view, more systematic studies are needed to evaluate the long-term stability and reproducibility of these biosensors under real environmental conditions. In parallel, the development of sustainable, environmentally friendly synthesis routes for electrode modifiers is essential to align sensor fabrication with green chemistry principles. Lastly, the automation of electrochemical measurement platforms represents a promising frontier to improve throughput, reproducibility, and accessibility for real-world environmental monitoring.

Taken together, we believe that, the progress summarized herein provides a strong foundation for future innovations in electrochemical sensing of pharmaceutical contaminants, moving the field closer to practical, and ecologically responsible monitoring solutions.

CRediT authorship contribution statement

Xhensiana Ndreka: Writing – original draft, Methodology, Investigation. **Jiří Kudr:** Writing – review & editing, Supervision, Conceptualization. **Ludovico Migliaccio:** Writing – review & editing, Supervision. **Ondřej Zítka:** Visualization, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No new data were generated or analysed as part of this review.

References

[1] T. aus der Beek, et al., Pharmaceuticals in the environment–global occurrences and perspectives, Environ. Toxicol. Chem. 35 (4) (2016) 823–835.
[2] D.L. S, B. V.G, V. Murali, From prescription to pollution: the ecological consequences of NSAIDs in aquatic ecosystems, Toxicol. Rep. 13 (2024) 101775.
[3] J. Žur, et al., Organic micropollutants paracetamol and ibuprofen-toxicity, biodegradation, and genetic background of their utilization by bacteria, Environ. Sci. Pollut. Res. Int. 25 (22) (2018) 21498–21524.
[4] B. Nunes, et al., Toxic potential of paracetamol to freshwater organisms: a headache to environmental regulators? Ecotoxicol. Environ. Saf. 107 (2014) 178–185.
[5] ChemAnalyst, Paracetamol Market Analysis: Industry Market Size, Plant Capavity, Production, Operating Effiency, Demand&Supply Gap, End-Use Applications, Distribution Channel, Regional Demand, Company Share, Manufacturing Process, 2015-2034, March 2024.
[6] ChemAnalyst, Ibuprofen Market Analysis: Industry Market Size, Plant Capavity, Production, Operating Effiency, Demand&Supply Gap, End-Use Applications, Distribution Channel, Regional Demand, Company Share, Manufacturing Process, 2015-2034, March 2024.
[7] T. Boogaerts, et al., Current and future perspectives for wastewater-based epidemiology as a monitoring tool for pharmaceutical use, Sci. Total. Environ. 789 (2021) 148047.
[8] N. Farhadi, et al., Ibuprofen elimination from water and wastewater using sonication/ultraviolet/hydrogen peroxide/zeolite-titanate photocatalyst system, Environ. Res. 198 (2021) 111260.
[9] A. Al-kaf, et al., Occurrence of Paracetamol in Aquatic Environments and Transformation by Microorganisms: A Review (2017) 1.
[10] S. Sharma, R. Yadav, A. Singh, Presence of medicinal materials in drinking water: a review, Mater. Today: Proc. (2021) 61.
[11] A. yamindago, et al., Sublethal effects of acetaminophen exposure on benthic aquatic animal (Hydra magnipapillata), Mol. Cell Toxicol. (2023) 20.
[12] S. Das, et al., IBUPROFEN : Its toxic effect on aquatic organisms (2022).
[13] P.J. Wiffen, et al., Oral paracetamol (acetaminophen) for cancer pain, Cochrane Database Syst. Rev. 7 (7) (2017).
[14] J. Alchin, et al., Why paracetamol (acetaminophen) is a suitable first choice for treating mild to moderate acute pain in adults with liver, kidney or cardiovascular disease, gastrointestinal disorders, asthma, or who are older, Curr. Med. Res. Opin. 38 (2022) 1–36.
[15] M.M. Rahman, et al., Modification of ibuprofen to improve the medicinal effect; structural, biological, and toxicological study, Heliyon. 10 (5) (2024) e27371.
[16] A. Sibiya, et al., Response of hepatic biochemical parameters and neurotoxicity to carbamazepine and ibuprofen in Oreochromis mossambicus, Fish. Physiol. Biochem. 49 (2023) 1–13.
[17] E. Tiedeken, et al., Monitoring, sources, receptors, and control measures for three European Union watch list substances of emerging concern in receiving waters – A 20 year systematic review, Sci. Total Environ. (2017) 574.
[18] T. Austin, et al., Ibuprofen exposure in Europe; ePiE as an alternative to costly environmental monitoring, Environ. Res. 209 (2022) 112777.
[19] S. Fatima, et al., Toxicity of NSAID drug (paracetamol) to nontarget organism-nostoc muscorum, Environ. Sci. Pollut. Res. Int. 27 (28) (2020) 35208–35216.
[20] J.-l. Wu, et al., Occurrence, removal and risk evaluation of ibuprofen and acetaminophen in municipal wastewater treatment plants: a critical review, Sci. Total Environ. 891 (2023) 164600.
[21] T.-H. Yu, et al., Biodegradation and bio-sorption of antibiotics and non-steroidal anti-inflammatory drugs using immobilized cell process, Chemosphere 84 (9) (2011) 1216–1222.
[22] V.K. Parida, et al., Emerging contaminants in wastewater: a critical review on occurrence, existing legislations, risk assessment, and sustainable treatment alternatives, J. Environ. Chem. Eng. 9 (5) (2021) 105966.
[23] L. Feng, et al., Removal of residual anti-inflammatory and analgesic pharmaceuticals from aqueous systems by electrochemical advanced oxidation processes. A review, Chem. Eng. J. 228 (2013) 944–964.
[24] M.D.G. de Luna, et al., Acetaminophen degradation by electro-fenton and photoelectro-fenton using a double cathode electrochemical cell, J. Hazard. Mater. 217-218 (2012) 200–207.
[25] F. Méndez-Arriaga, et al., Ultrasonic treatment of water contaminated with ibuprofen, Water. Res. 42 (16) (2008) 4243–4248.
[26] M. Negarestani, et al., Simultaneous removal of acetaminophen and ibuprofen from underground water by an electrocoagulation unit: operational parameters and kinetics, Groundw. Sustain. Dev. 11 (2020) 100474.

[27] F.P. Shariati, et al., Membrane bioreactor for treatment of pharmaceutical wastewater containing acetaminophen, Desalination. 250 (2) (2010) 798–800.
[28] G. Crini, E. Lichtfouse, Advantages and disadvantages of techniques used for wastewater treatment, Environ. Chem. Lett. 17 (1) (2019) 145–155.
[29] J. Fernandes, P.J. Ramísio, H. Puga, A comprehensive review on various phases of wastewater technologies: trends and future perspectives, Eng 5 (2024) 2633–2661, https://doi.org/10.3390/eng5040138.
[30] A.I. Osman, et al., Advanced adsorbents for ibuprofen removal from aquatic environments: a review, Environ. Chem. Lett. 22 (1) (2024) 373–418.
[31] M. López Zavala, C.R. Jaber Lara, Degradation of paracetamol and its oxidation products in surface water by electrochemical oxidation, Environ. Eng. Sci. 35 (11) (2018) 1248–1254.
[32] Y. Vicente-Martínez, M. Caravaca, M. Briceño, Spectrophotometric nanodrop system for quantification of trace concentrations of ibuprofen in water samples using silver-functionalized magnetic nanoparticles, Microchem. J. 180 (2022) 107555.
[33] N. Rahman, F.K. Omar, HPLC method for determination of paracetamol in pharmaceutical formulations and environmental water samples (2018).
[34] M. Pedrouzo, et al., Pharmaceutical determination in surface and wastewaters using high-performance liquid chromatography-(Electrospray)-mass spectrometry, J. Sep. Sci. 30 (2007) 297–303.
[35] M. Pedrouzo, et al., Pharmaceutical determination in surface and wastewaters using high-performance liquid chromatography-(electrospray)-mass spectrometry, J. Sep. Sci. 30 (3) (2007) 297–303.
[36] T. Qureshi, et al., Determination of Ibuprofen Drug in Aqueous Environmental Samples by Gas Chromatography–mass Spectrometry without Derivatization 1 (2014) 45–54.
[37] M. Esterhuizen-Londt, et al., LC-MS/MS method development for quantitative analysis of acetaminophen uptake by the aquatic fungus Mucor hiemalis, Ecotoxicol. Environ. Saf. 128 (2016) 230–235.
[38] J. Noiphung, et al., Nanomaterials-based electrochemical sensors and biosensors for the detection of non-steroidal anti-inflammatory drugs, TrAC Trends Anal. Chem. 143 (2021) 116403.
[39] T.S.S. Naik, et al., Poly (L-leucine) modified carbon paste electrode as an electrochemical sensor for the detection of paracetamol in presence of folic acid, Mater. Sci. Energy Technol. (2020) 3.
[40] B. Feier, et al., Electrochemical detection and removal of pharmaceuticals in waste waters, Curr. Opin. Electrochem. 11 (2018) 1–11.
[41] X. Ndreka, et al., Electrochemical determination of wine polyphenols using carbon electrodes: a review, Electroanalysis. 36 (8) (2024) e202400084.
[42] N. Isak, et al., Rapid and sensitive electrochemical determination of flavonoids in Albanian wines using zeolite X and Prrenjasi clay as carbon paste modifiers, J Electrochem. Sci. Eng. (2025) 2682.
[43] D. González-Morales, et al., Development of a low-cost UV-vis spectrophotometer and its application for the detection of mercuric ions assisted by chemosensors, Sensors 20 (2020), https://doi.org/10.3390/s20030906.
[44] UHPLCs, *Do you really know how much does an HPLC system cost ?* https://uhplcs.com/do-you-really-know-how-much-does-an-hplc-system-cost/.
[45] LabX, *The best mass spectrometers: a buyer's review of price and features.* https://www.labx.com/resources/the-best-mass-spectrometers-a-buyers-review-of-price-and-features/4808.
[46] Supplies, M., *How much does a potentiostat cost?* https://www.msesupplies.com/pages/search-results?q=potentiostat.
[47] L.M.A. Melo, et al., Electrochemical methods for the determination of acetaminophen in biological matrices: a critical review in the clinical field, Anal. Chim. Acta 1333 (2025) 343243.
[48] M. Stoytcheva, et al., Advances in electrochemical sensors for paracetamol detection: electrode materials, modifications, and analytical applications, Int. J. Electrochem. Sci. 20 (2) (2025) 100924.
[49] H. Ouafy, et al., Determination of Ibuprofen in tablets and Human blood samples: effect of clay in minimizing overvoltage of ibuprofen oxidation, ChemistrySelect. 4 (2019) 11282–11287.
[50] U. Freo, et al., Paracetamol: a review of guideline recommendations, J. Clin. Med. (15) (2021) 10.
[51] S. Motov, et al., Comparison of oral ibuprofen at three single-dose regimens for treating acute pain in the Emergency department: a randomized controlled trial, Ann. Emerg. Med. 74 (4) (2019) 530–537.
[52] S. Webb, et al., Indirect human exposure to pharmaceuticals via drinking water, Toxicol. Lett. 142 (3) (2003) 157–167.
[53] F. Cabrera-Diaz, et al., Implementation and outcome of an electronic tool for detection of paracetamol overdose in a tertiary care hospital, Int. J. Clin. Pharm. (2020) 43.
[54] B. Park, D. Antoine, Paracetamol (acetaminophen) poisoning, BMJ Clin. Evid. (2015), 2015.
[55] A.A. Somani, et al., Evaluation of changes in oral drug absorption in preterm and term neonates for Biopharmaceutics Classification System (BCS) class I and II compounds, Br. J. Clin. Pharmacol. 81 (1) (2016) 137–147.
[56] T. J, in: V.B. Patel, V.R. Preedy (Eds.), *Chapter 45 - Curcumin, antioxidant activity, and Paracetamol toxicity*, in *Toxicology*, Academic Press, 2021, pp. 469–477.
[57] T. Roušar, et al., Purified acetaminophen-glutathione conjugate is able to induce oxidative stress in rat liver mitochondria, Physiol. Res. 61 (Suppl 2) (2012) S103–S109.
[58] K. Allegaert, J.N. van den Anker, Perinatal and neonatal use of paracetamol for pain relief, Semi. in Fetal Neonat. Med. 22 (5) (2017) 308–313.
[59] P. Mészáros, et al., Investigation of intestinal absorption and excretion of paracetamol in streptozotocin-induced hyperglycemia, Int. J. Mol. Sci. 23 (2022), https://doi.org/10.3390/ijms231911913.
[60] L. Sahyoun, N. Pyrsopoulos, Liver injury induced by paracetamol and challenges associated with intentional and unintentional use, World J. Hepatol. 12 (2020) 125–136.
[61] C. Alvarez, et al., Investigation on the possibility of biowaivers for Ibuprofen, J. Pharm. Sci. 100 (2011) 2343–2349.
[62] S. Tiwari, et al., Gender-specific changes in energy metabolism and protein degradation as major pathways affected in livers of mice treated with ibuprofen, Sci. Rep. 10 (1) (2020) 3386.
[63] J. Millecam, et al., In vivo metabolism of ibuprofen in growing conventional pigs: a pharmacokinetic approach, Front. Pharmacol. (2019) 10.
[64] A. Gliszczyńska, E. Sánchez-López, Dexibuprofen therapeutic advances: prodrugs and nanotechnological formulations, Pharmaceutics. 13 (2021), https://doi.org/10.3390/pharmaceutics13030414.
[65] R. Monrad, et al., Dissecting the reaction of phase II metabolites of ibuprofen and other NSAIDS with human plasma protein, Chem. Sci. (2014) 5.
[66] B. Healy, et al., Electrochemical determination of acetaminophen at a carbon electrode modified in the presence of β-cyclodextrin: role of the activated glassy carbon and the electropolymerised β-cyclodextrin, J. Solid State Electrochem. 25 (10) (2021) 2599–2609.
[67] G. Filho, et al., Controlled release of drugs from cellulose acetate matrices produced from sugarcane bagasse: monitoring by square-wave voltammetry, Drug Dev. Ind. Pharm. 42 (2015) 1–7.
[68] J.D. Sawant, K.K. Patil, G.S. Gokavi, Kinetics and mechanism of oxidation of paracetamol by an Anderson-type 6-molybdocobaltate(III) in acidic medium, Trans. Metal Chem. 44 (2) (2019) 153–159.
[69] Q. Qin, X. Bai, Z. Hua, Electropolymerization of a conductive β-cyclodextrin polymer on reduced graphene oxide modified screen-printed electrode for simultaneous determination of ascorbic acid, dopamine and uric acid, J. Electroanal. Chem. 782 (2016) 50–58.
[70] S.K. Hassaninejad-Darzi, F. Shajie, Simultaneous determination of acetaminophen, pramipexole and carbamazepine by ZSM-5 nanozeolite and TiO (2) nanoparticles modified carbon paste electrode, Mater. Sci. Eng. C. Mater. Biol. Appl. 91 (2018) 64–77.
[71] M. Carolyne Prete, et al., Electrochemical determination of ibuprofen by batch-injection analysis using a BORON-doped ultrananocrystalline diamond electrode, Electroanalysis. 37 (1) (2025) e202400121.
[72] S. Amin, et al., Disposable screen printed graphite electrode for the direct electrochemical determination of ibuprofen in surface water, Environ. Nanotechnol. Monit. Manage 1-2 (2014) 8–13.
[73] Chemistry, O.a.R.i.O., *LibreTexts Chem*.
[74] A. Lima, et al., Electrochemical oxidation of Ibuprofen and its voltammetric determination at a boron-doped diamond electrode, Electroanalysis. (2013) 25.
[75] I. Apetrei, et al., Determination of ibuprofen based on screen-printed electrodes modified with carbon nanofibers, Farmacia 65 (2017) 790–795.
[76] M.T. Soomro, et al., A simple electrochemical approach for determination and direct monitoring of drug degradation in water, Curr. World Environ. J. 8 (2013) 375–380.
[77] A.B. Lima, et al., Simultaneous determination of paracetamol and ibuprofen in pharmaceutical samples by differential pulse voltammetry using a boron-doped diamond electrode, J. Braz. Chem. Soc. (2014) 25.
[78] J. Stefano, et al., Flow-injection analysis with multiple-pulse amperometry for simultaneous determination of paracetamol and naproxen using a homemade Flow cell for screen-printed electrodes, J. Braz. Chem. Soc. 25 (2014) 484–491.
[79] M. Stoytcheva, et al., A zirconia-nanoparticles-modified carbon paste electrode for voltammetric determination of ibuprofen, ChemistrySelect. 7 (48) (2022) e202203950.
[80] B.-R. Adhikari, M. Govindhan, A. Chen, Carbon nanomaterials based electrochemical sensors/biosensors for the sensitive detection of pharmaceutical and biological compounds, Sensors 15 (2015) 22490–22508, https://doi.org/10.3390/s150922490.
[81] L. Taouri, et al., A new highly sensitive micro-sensor for the ultra-traces analysis of paracetamol directly in water, Microchem. J. 178 (2022) 107380.
[82] F.C. Vicentini, et al., Imparting improvements in electrochemical sensors: evaluation of different carbon blacks that give rise to significant improvement in the performance of electroanalytical sensing platforms, Electrochim. Acta 157 (2015) 125–133.
[83] S. Jebril, et al., A novel electrochemical sensor modified with green gold sononanoparticles and carbon black nanocomposite for bisphenol A detection, Mater. Sci. Eng.: B 264 (2021) 114951.
[84] L. Gullo, et al., Carbon black as a cost-effective nanomodifier for (electro) chemical-free pre-treatment thermoplastic polyurethane-based 3D printed electrodes, Electrochim. Acta 482 (2024) 143982.
[85] G. Nalon, et al., Effects of different kinds of carbon black nanoparticles on the piezoresistive and mechanical properties of cement-based composites, J. Build. Eng. 32 (2020) 101724.
[86] A. Aboalhassan, et al., Controlled alignment of carbon black nanoparticles in electrospun carbon nanofibers for flexible films, Polymers. (Basel) 16 (2024), https://doi.org/10.3390/polym16030327.
[87] S. Wu, L. Zhang, J. Chen, Paracetamol in the environment and its degradation by microorganisms, Appl. Microbiol. Biotechnol. 96 (4) (2012) 875–884.
[88] W. Yi, et al., A strategy of electrochemical simultaneous detection of acetaminophen and levofloxacin in water based on g-C 3 N 4 nanosheet-doped graphene oxide, Environ. Sci.: Nano 8 (2020).

[89] P. Balasubramanian, et al., Sonochemical synthesis of molybdenum oxide (MoO3) microspheres anchored graphitic carbon nitride (g-C3N4) ultrathin sheets for enhanced electrochemical sensing of Furazolidone, Ultrason. Sonochem. 50 (2019) 96–104.

[90] Y. Huang, et al., Synthesis of CuO/g-C3N4 composites, and their application to voltammetric sensing of glucose and dopamine, Microchimica Acta (2018) 186.

[91] N. Murugan, M. Chan, A. Sundramoorthy, Electrochemical detection of uric acid on exfoliated nanosheets of graphitic-like carbon nitride (g-C 3 N 4) based sensor, J. Electrochem. Soc. 166 (2019) B3163–B3170.

[92] C.-Z. Li, et al., Graphitic carbon nitride nanosheets coated carbon black as high-performance PtRu catalyst support material for methanol electrooxidation, J. Mater. Chem. A 2 (2014).

[93] Q. Lu, et al., One-step electrochemical synthesis of ultrathin graphitic carbon nitride nanosheets and their application to the detection of uric acid, Chem. Commun. (Camb) 51 (61) (2015) 12251–12253.

[94] H. Zhang, et al., Graphitic carbon nitride nanosheets doped graphene oxide for electrochemical simultaneous determination of ascorbic acid, dopamine and uric acid, Electrochim. Acta 142 (2014) 125–131.

[95] A. Sasal, et al., Direct determination of paracetamol in environmental samples using screen-printed carbon/carbon nanofibers sensor - experimental and theoretical studies, Electroanalysis. (2020) 32.

[96] T.-W. Tseng, et al., Construction of strontium phosphate/graphitic-carbon nitride: a flexible and disposable strip for acetaminophen detection, J. Hazard. Mater. 410 (2021) 124542.

[97] M.Z. Iqbal, et al., Synergetic electrochemical performance of strontium phosphate/polyaniline/graphene for high performance supercapattery devices, Diam. Relat. Mater. 124 (2022) 108918.

[98] M. Parvinzadeh Gashti, M. Stir, J. Hulliger, Growth of strontium hydrogen phosphate/gelatin composites: a biomimetic approach, N. J. Chem. 40 (2016) 5495–5500.

[99] N. Serrano, et al., Commercial screen-printed electrodes based on carbon nanomaterials for a fast and cost-effective voltammetric determination of paracetamol, ibuprofen and caffeine in water samples, Sensors 19 (2019), https://doi.org/10.3390/s19184039.

[100] C. Nnodum, K. Yusuf, Ecological risk assessment of pharmaceutical residues in surface water, Int. J. Sci. Res. Manage. 10 (2022) 56–62.

[101] S. Motoc, et al., Highly sensitive and simultaneous electrochemical determinations of non-steroidal anti-inflammatory drugs in water using nanostructured carbon-based paste electrodes, Sci. Total Environ. 846 (2022) 157412.

[102] H. Gergeroglu, S. Yildirim, M.F. Ebeoglugil, Nano-carbons in biosensor applications: an overview of carbon nanotubes (CNTs) and fullerenes (C60), SN. Appl. Sci. 2 (4) (2020) 603.

[103] A. Power, et al., Carbon nanomaterials and their application to electrochemical sensors; A review, Nanotechnol. Rev. (2017) 7.

[104] C. Gibi, et al., Carbon materials for electrochemical sensing application – A mini review, J. Taiwan. Inst. Chem. Eng. 154 (2024) 105071.

[105] X. Luo, et al., Application of nanoparticles in electrochemical sensors and biosensors, Electroanalysis. 18 (4) (2006) 319–326.

[106] S. Weheabby, et al., Paracetamol detection in environmental and pharmaceutical samples using multi-walled carbon nanotubes decorated with silver nanoparticles, Microchem. J. 193 (2023) 109192.

[107] Z. Wang, et al., Synthesis of silver nanoparticle/multi-walled carbon nanotube composites and their application in electronic pastes, Nanomaterials 15 (2025), https://doi.org/10.3390/nano15030152.

[108] B. Habibi, M. Jahanbakhshi, Silver nanoparticles/multi walled carbon nanotubes nanocomposite modified electrode: voltammetric determination of clonazepam, Electrochim. Acta 118 (2014) 10–17.

[109] I. Anshori, et al., Functionalized multi-walled carbon nanotube/silver nanoparticle (f-MWCNT/AgNP) nanocomposites as non-enzymatic electrochemical biosensors for dopamine detection, Nanocomposites 7 (1) (2021) 97–108.

[110] T.M.B.F. Oliveira, S. Morais, New generation of electrochemical sensors based on multi-walled carbon nanotubes, Appl. Sci. 8 (2018), https://doi.org/10.3390/app8101925.

[111] S.K. Adhikary, Ž. Rudžionis, R. Rajapriya, The effect of carbon nanotubes on the flowability, mechanical, microstructural and durability properties of cementitious composite: an overview, Sustainability. 12 (2020), https://doi.org/10.3390/su12208362.

[112] W.-J. Zeng, et al., Carboxylated multi-walled carbon nanotubes (MWCNTs-COOH)-intercalated graphene oxide membranes for highly efficient treatment of organic wastewater, J. Water. Process. Eng. 40 (2021) 101901.

[113] T. Kupka, et al., Carbon nanotubes in cement—A new approach for building composites and its influence on environmental effect of material, Molecules. 29 (2024), https://doi.org/10.3390/molecules29225379.

[114] R. Gupta, B. Singh, Chemical modification of carboxylated MWCNTs for enhanced electrical conducting and magnetic properties, Mater. Sci. Eng.: B 262 (2020) 114730.

[115] M. Zahran, et al., Recent advances in silver nanoparticle-based electrochemical sensors for determining organic pollutants in water: a review, Mater. Adv. 2 (22) (2021) 7350–7365.

[116] R. Bala, et al., A supersensitive silver nanoprobe based aptasensor for low cost detection of malathion residues in water and food samples, Spectrochimica Acta Part A: Mol. Biomol. Spectroscopy 196 (2018) 268–273.

[117] M. Haššo, et al., Proving the automatic benchtop electrochemical station for the development of dopamine and paracetamol sensors, Microchimica Acta 191 (7) (2024) 408.

[118] G. Paimard, E. Ghasali, M. Baeza, Screen-printed electrodes: fabrication, modification, and biosensing applications, Chemosensors (2023) 11, https://doi.org/10.3390/chemosensors11020113.

[119] D. Antuña-Jiménez, et al., Screen-printed electrodes modified with metal nanoparticles for small molecule sensing, Biosensors. (Basel) 10 (2020), https://doi.org/10.3390/bios10020009.

[120] A.M. Alkhudaydi, et al., Novel sensitive electrochemical detection of paracetamol using magnetite/MXene electrode by differential pulse voltammetry, Int. J. Electrochem. Sci. (2024).

[121] W. Li, et al., Three dimensional nanosuperstructures made of two-dimensional materials by design: synthesis, properties, and applications, Nano Today 29 (2019) 100799.

[122] Y. Zhang, et al., Simultaneous voltammetric determination of acetaminophen and isoniazid using MXene modified screen-printed electrode, Biosens. Bioelectron. 130 (2019) 315–321.

[123] Z. Li, et al., Synthesis and thermal stability of two-dimensional carbide MXene Ti3C2, Mater. Sci. Eng.: B (2014) 191.

[124] K. Rizwan, et al., MXene-based electrochemical and biosensing platforms to detect toxic elements and pesticides pollutants from environmental matrices, Chemosphere 291 (2022) 132820.

[125] X. Li, et al., Applications of MXene (Ti3C2Tx) in photocatalysis: a review, Mater. Adv. 2 (5) (2021) 1570–1594.

[126] C. Ferrara, et al., Ti3C2Tx MXene compounds for electrochemical energy storage, Curr. Opin. Electrochem. 29 (2021) 100764.

[127] T. Wang, L. Zhu, H. Kanda, Ti3C2 MXene-TiO2 hybrid-modified U-bend fiberoptic sensor for improved refractive index sensitivity and ammonia detection, Sens. Actuat. B: Chem. 393 (2023) 134136.

[128] E. Gaya, et al., Fe3O4-Nanoparticle-modified sensor for the detection of dopamine, uric acid and ascorbic acid, Chemosensors (2023) 11, https://doi.org/10.3390/chemosensors11020079.

[129] M. Nguyen, et al., Fe3O4 Nanoparticles: structures, synthesis, magnetic properties, surface functionalization, and emerging applications, Appl. Sci. 11 (2021) 11301.

[130] J. Kudr, et al., Magnetic solids in electrochemical analysis, TrAC Trends Anal. Chem. 98 (2018) 104–113.

[131] K. Akhtar, et al., Phytoextract based synthesis of TiO2.Al2O3 nanocomposites for efficient electrocatalytic detection of acetaminophen from environmental and pharmaceutical samples, Ceram. Int. 50 (7) (2024) 11012–11021. Part A.

[132] S. Mansoor, et al., Green synthesis of a MnO-GO-Ag nanocomposite using leaf extract of fagonia arabica and its antioxidant and anti-inflammatory performance, Nano-Struct. Nano-Obj. 29 (2022) 100835.

[133] H. Zulfaqar, et al., Synthesis of silver nanoparticles using Fagonia Cretica and their antimicrobial activities, Nanoscale Adv. (2019) 1.

[134] S. Shahid, et al., CuO-GO-Ag; Green synthesis with Fagonia Arabica and Biomedical Potential is a bioinspired nano theranostics composite, Front. Mater. 9 (2022) 875148.

[135] A. Kader Mohiuddin, M.S. Ahmed, S. Jeon, Palladium doped α-MnO2 nanorods on graphene as an electrochemical sensor for simultaneous determination of dopamine and paracetamol, Appl. Surf. Sci. 578 (2022) 152090.

[136] L.M. Housel, et al., Investigation of α-MnO2 tunneled structures as model cation hosts for energy storage, Acc. Chem. Res. 51 (3) (2018) 575–582.

[137] C. Zhu, et al., Novel electrochemical sensor based on MnO2 nanowire modified carbon paper electrode for sensitive determination of tetrabromobisphenol A, Chemosensors (2023) 11, https://doi.org/10.3390/chemosensors11090482.

[138] Y.Z. Weikai Xiang, Zheng Jiang, Xiaopeng Li, Hao Zhang, Yu Sun, Zhijun Ning, Fuping Du, Peng Gao, Jing Qian, Kenichi Kato, Miho Yamauchi, Yuhan Sun, Palladium single atoms supported by interwoven carbon nanotube and manganese oxide nanowire networks for enhanced electrocatalysis, J. Mater. Chem A 6 (46) (2018) 23366–23377.

[139] J. Kang, et al., Simultaneous voltammetric detection of acetaminophen and levofloxacin based on barley leaves derived carbon decorated with gold nanoparticles, Microchem. J. 205 (2024) 111335.

[140] H. Zhang, et al., Biomass-derived carbon materials for electrochemical sensing: recent advances and future perspectives, Crit. Rev. Anal. Chem. (2024) 1–26.

[141] L. Wang, et al., Electrochemical sensing and biosensing platform based on biomass-derived macroporous carbon materials, Anal. Chem. 86 (3) (2014) 1414–1421.

[142] M. Fan, et al., Preparation of activated carbon with recycled ZnCl2 for maximizing utilization efficiency of the activating agent and minimizing generation of liquid waste, Chem. Eng. J. 500 (2024) 157278.

[143] A. Ouakkaf, et al., Activated Carbon from Biomass from Zncl2 Activation for Methylene Blue Adsorption (2023).

[144] Lizzy Mokaba, et al., Electro-oxidation of ibuprofen and metoprolol using a Manganese oxide platform, ChemistrySelect. 9 (38) (2024) e202400385.

[145] N. Kajal, et al., Metal organic frameworks for electrochemical sensor applications: a review, Environ. Res. 204 (2022) 112320.

[146] F. Zhao, Metal-organic frameworks-based electrochemical sensors and biosensors, Int. J. Electrochem. Sci. 14 (2019) 5287–5304.

[147] S. Yuan, et al., Stable metal–Organic frameworks: design, synthesis, and applications, Adv. Mater. (37) (2018) 30.

[148] A. Kidanemariam, S. Cho, Recent advances in the application of metal–Organic frameworks and coordination polymers in electrochemical biosensors, Chemosensors 12 (2024), https://doi.org/10.3390/chemosensors12070135.

[149] A. Karmakar, E. Velasco, J. Li, Metal-organic frameworks as effective sensors and scavengers for toxic environmental pollutants, Natl. Sci. Rev. 9 (7) (2022).
[150] X. Fang, B. Zong, S. Mao, Metal-organic framework-based sensors for environmental contaminant sensing, Nanomicro Lett. 10 (4) (2018) 64.
[151] M.K. Muthukumaran, et al., Recent advances in metal-organic frameworks for electrochemical sensing applications, Talanta Open. 11 (2025) 100396.
[152] K. Theyagarajan, B.A. Lakshmi, Y.-J. Kim, Electrochemical sensing of acetaminophen in biofluids, pharmaceutical and environmental samples using cobalt hexacyanoferrate decorated iron terephthalate metal organic framework, Electrochim. Acta 488 (2024) 144229.
[153] K. Yang, et al., Electrochemical sensor based on cobalt single-atom anchored porous carbon composite for sensitive detection of acetaminophen, Microchem. J. 203 (2024) 110874.
[154] Y. Guan, et al., Fabrication of Cu-BTC@PW12/GO for multivariate sensing dopamine and acetaminophen as electrochemical sensor, J. Environ. Chem. Eng. 12 (5) (2024) 114078.
[155] A. Mathai Varghese, et al., Enhancing effect of UV activation of graphene oxide on carbon capture performance of metal-Organic Framework /graphene oxide hybrid adsorbents, Chem. Eng. J. 420 (2021) 129677.
[156] L. Ji, et al., Solvent-exfoliated Cu-TCPP nanosheets: electrochemistry and sensing application in simultaneous determination of 4-aminophenol and acetaminophen, Microchem. J. 181 (2022) 107688.
[157] X. Xu, et al., Construction of electrochemical and photoelectrochemical sensing platform based on porphyrinic metal-organic frameworks for determination of ascorbic acid, Nanomaterials 12 (2022), https://doi.org/10.3390/nano12030482.
[158] B. Li, Ultrathin Cu-TCPP MOF nanosheets: a new theragnostic nanoplatform with magnetic resonance/near-infrared thermal imaging for synergistic phototherapy of cancers, Theranostics. 8 (2018) 4086–4096.
[159] L.-J. Han, et al., A highly solvent-stable metal–Organic framework nanosheet: morphology control, exfoliation, and luminescent property, Small. 14 (17) (2018) 1703873.
[160] L. Ji, et al., Ultrasensitive and simple dopamine electrochemical sensor based on the synergistic effect of Cu-TCPP frameworks and graphene nanosheets, Molecules. (2023) 28, https://doi.org/10.3390/molecules28062687.
[161] Y. Dong, et al., Construction of bimetallic MOF derived nickel–cobalt selenide on carbon cloth as integrated electrode for acetaminophen sensing, Microchem. J. 208 (2025) 112400.
[162] Y. Xing, et al., Three-dimensional Setaria viridis-like NiCoSe2 nanoneedles array: as an efficient electrochemical hydrazine sensor, Colloids Surfaces A: Physicochem. Eng. Aspects 650 (2022) 129549.
[163] J. Yu, et al., Metallic and superhydrophilic nickel cobalt diselenides nanosheets electrodeposited on carbon cloth as bifunctional electrocatalyst, J. Mater. Chem. A (2018) 6.
[164] A. Xie, et al., Electrochemical sensor based on ZIF-8/67 derived ZnCo2O4@Fe3O4/MWCNTs with core–shell structure for simultaneous detection of paraminophen and paracetamol, Microchem. J. 204 (2024) 111101.
[165] Y. Dong, M. Zhou, L. Zhang, 3D multiporous Co,N co-doped MoO2/MoC nanorods hybrids as improved electrode materials for highly sensitive simultaneous determination of acetaminophen and 4-aminophenol, Electrochim. Acta 302 (2019) 56–64.
[166] H. Guo, et al., The effect of Co and N of porous carbon-based materials fabricated via sacrificial templates MOFs on improving DA and UA electrochemical detection, Microporous Mesoporous Mater. 263 (2018) 21–27.
[167] J. Guo, et al., Electrochemical sensing for simultaneous determination of dopamine and acetaminophen based on ZIF-8@ZIF-67 derived Co/Mo2C embedded in N-doped carbon hollow nanocages, Microchem. J. 206 (2024) 111642.
[168] J. Gu, et al., A dual-functional and efficient MOF-5@MWCNTs electrochemical sensing device for the measurement of trace-level acetaminophenol and dopamine, Molecules. 29 (2024), https://doi.org/10.3390/molecules29235534.
[169] Á. Terán-Alcocer, et al., Electrochemical sensors based on conducting polymers for the aqueous detection of biologically relevant molecules, Nanomaterials (2021) 11, https://doi.org/10.3390/nano11010252.
[170] A. Pollap, et al., Electrochemical sensing of ciprofloxacin and paracetamol in environmental water using titanium sol based sensor, J. Electroanal. Chem. 878 (2020) 114574.
[171] Y. Jeong, et al., Development of modified mesoporous carbon (CMK-3) for improved adsorption of bisphenol-A, Chemosphere 238 (2020) 124559.
[172] P. Xie, et al., A sensitive electrochemical sensor based on wrinkled mesoporous carbon nanomaterials for rapid and reliable assay of 17β-estradiol, Electrochim. Acta 408 (2022) 139960.
[173] M. Regiart, et al., An ordered mesoporous carbon modified electrochemical sensor for solid-phase microextraction and determination of triclosan in environmental samples, Sens. Actuat. B: Chem. 232 (2016) 765–772.
[174] H. Wang, et al., Ultrasensitive electrochemical sensor for determination of trace carbadox with ordered mesoporous carbon/GCE, J. Electroanal. Chem. 857 (2020) 113736.
[175] S. Sekar, J.-S. Yun, S. Lee, Metal-free electrocatalytic nanocomposites of poly azovan blue-decorated graphitic carbon nitride for simultaneously sensing paracetamol and 4-aminophenol, Environ. Res. 239 (2023) 117293.
[176] A. Porfireva, et al., Electrochemical sensor based on poly(Azure B)-DNA composite for doxorubicin determination, Sensors 19 (2019), https://doi.org/10.3390/s19092085.
[177] D. Shan, et al., A composite poly azure B /clay /enzyme sensor for the mediated electrochemical determination of phenols, J. Electroanal. Chem. 537 (2002) 103–109.
[178] B. Yuan, et al., Nitrogen and sulfur co-doped porous carbon with metal–organic framework derived Co3O4@NiCo2O4 core–shell nanocomposites for simultaneous detection of mesalazine and acetaminophen, Microchem. J. 208 (2025) 112633.
[179] X. Gong, et al., A review on zeolitic imidazolate framework-8 based materials with special wettability for oil/water separation, J. Environ. Chem. Eng. 11 (6) (2023) 111360.
[180] C. Zhang, et al., Design of rod-like NiCo2O4 with nanostructure derived from ZIF-67 as excellent material for supercapacitors, Int. J. Electrochem. Sci. 16 (4) (2021) 210455.
[181] S. Lv, et al., Nitrogen and sulfur co-doped porous chitosan hydrogel-derived carbons for supercapacitors, J. Electroanal. Chem. 907 (2022) 116060.
[182] H. Guo, et al., Simultaneous determination of 4-aminophenol and acetaminophen based on high electrochemical performance of ZIF-67/MWCNT-COOH/Nafion composite, Microchem. J. 158 (2020) 105262.
[183] G.I. Mohammed, et al., Electrochemical sensor for trace determination of timolol maleate drug in real samples and drug residues using Nafion/carboxylated-MWCNTs nanocomposite modified glassy carbon electrode, Microchem. J. 143 (2018) 474–483.
[184] S. Yang, et al., Voltammetric determination of theophylline at a nafion/multi-wall carbon nanotubes composite film-modified glassy carbon electrode, J. Chem. Sci. 122 (6) (2010) 919–926.
[185] M. Shalauddin, et al., A metal free nanosensor based on nanocellulose-polypyrrole matrix and single-walled carbon nanotube: experimental study and electroanalytical application for determination of paracetamol and ciprofloxacin, Environ. Nanotechnol. Monit. Manage 18 (2022) 100691.
[186] J. Kim, S. Yun, Z. Ounaies, Discovery of cellulose as a smart material, Macromolecules. 39 (12) (2006) 4202–4206.
[187] K. Nekoueian, et al., Advanced nanocellulose-based electrochemical sensor for tetracycline monitoring, Electrochim. Acta 500 (2024) 144639.
[188] L. Wang, X. Wu, Z. Xie, Determination of enrofloxacin and its metabolite ciprofloxacin by high performance capillary electrophoresis with end-column amperometric detection, J. Sep. Sci. 28 (11) (2005) 1143–1148.
[189] M. Sheraz, et al., Cellulose-based electrochemical sensors, Sensors (2025) 25, https://doi.org/10.3390/s25030645.
[190] R. Jain, N. Jadon, A. Pawaiya, Polypyrrole based next generation electrochemical sensors and biosensors: a review, TrAC Trends Anal. Chem. 97 (2017) 363–373.
[191] C. Esmaeili, et al., Synergy effect of nanocrystalline cellulose for the biosensing detection of glucose, Sensors 15 (2015) 24681–24697, https://doi.org/10.3390/s151024681.
[192] M. Sharma, A.K. Sharma, S.K. Shukla, Potentiometric sensing of ibuprofen over ferric oxide doped chitosan grafted polypyrrole-based electrode, Int. J. Biol. Macromol. 268 (2024) 131598.
[193] A.V. Bounegru, I. Bounegru, Chitosan-based electrochemical sensors for pharmaceuticals and clinical applications, Polymers. (Basel) 15 (2023), https://doi.org/10.3390/polym15173539.
[194] Z. Jin, et al., Unlocking the potential of chitosan in immunoassay sensor, Carbohydr. Polym. 350 (2025) 123024.
[195] Sarikokba, et al., Development of a novel sensor with high sensitivity for electroanalytical determination of bisphenol A based on chitosan-3-mercaptopropyl trimethoxysilane modified glassy carbon electrode, Microchem. J. 181 (2022) 107748.
[196] S.J. Malode, M. Ali Alshehri, N.P. Shetti, Nanomaterial-based electrochemical sensors for the detection of pharmaceutical drugs, Chemosensors 12 (2024), https://doi.org/10.3390/chemosensors12110234.
[197] M. Hashem, G. Ali, Chitosan Its Appl. Sens. Mater. (2023).
[198] C. Colin, et al., Easy cleaning plus stable activation of glassy carbon electrode surface by oxygen plasma, Bioelectrochemistry. 154 (2023) 108551.
[199] D. García García, et al., Impact of cleaning procedures on screen-printed gold electrodes performance for mutation detection, J. Appl. Electrochem. 55 (7) (2025) 1937–1946.
[200] D. Stan, et al., What is the optimal method for cleaning screen-printed electrodes? Processes 10 (2022) https://doi.org/10.3390/pr10040723.
[201] M. Herrera-Domínguez, et al., Detection of acetaminophen in groundwater by laccase-based amperometric biosensors using MoS2 modified carbon paper electrodes, Sensors 23 (2023), https://doi.org/10.3390/s23104633.
[202] A.V. Bounegru, et al., Electrochemical sensors and biosensors for the detection of pharmaceutical contaminants in natural waters—A comprehensive review, Chemosensors 13 (2025), https://doi.org/10.3390/chemosensors13020065.
[203] M. Roushani, F. Shahdost-fard, Covalent attachment of aptamer onto nanocomposite as a high performance electrochemical sensing platform: fabrication of an ultra-sensitive ibuprofen electrochemical aptasensor, Mater. Sci. Eng.: C 68 (2016) 128–135.
[204] S. Motoc, et al., Electrochemical selective and simultaneous detection of diclofenac and Ibuprofen in aqueous solution using HKUST-1 metal-Organic Framework-carbon nanofiber composite electrode, Sensors 16 (2016), https://doi.org/10.3390/s16101719.
[205] J. Zitka, et al., Fully automated station for testing, characterizing and modifying screen-printed electrodes, Anal. Methods 14 (39) (2022) 3824–3830.

[206] I. Oh, et al., The Electrolab: an open-source, modular platform for automated characterization of redox-active electrolytes, Device 1 (5) (2023) 100103.
[207] M.A. Pence, G. Hazen, J. Rodríguez-López, An automated electrochemistry platform for studying pH-dependent molecular electrocatalysis, Digit. Discov. 3 (9) (2024) 1812–1821.
[208] S. Dieckhöfer, W. Schuhmann, E. Ventosa, Accelerated electrochemical investigation of Li, ChemElectroChem. 8 (16) (2021) 3143–3149.